



TABLE OF CONTENTS

1 title page

2 table of contents

12/2025 architecture for autism - Rhino individual

3 design principles

4 Bambulab 3D printed model

5 nuerodivergent / neurotypical childcare center

6 public space

7 apartments with private / semi-privtae / semi-public space

11/2024 childcare center (CPE) - Revit individual

8 technical plans

9 exterior wall construction detail section

07/2024 les maisons enjouées (CPE) - Revit individual

10 professional work sample

04/2018 le plateau public library - Rhino individual

11 rendered perspective exterior

12 plans

13 axonometry

14 rendered perspectives interior

12/2019 rosemont public pool and cultural center - Rhino individual

15 concept diagram axonometry

16 exterior tectonics and energy design

17 exterior wall section construction detail

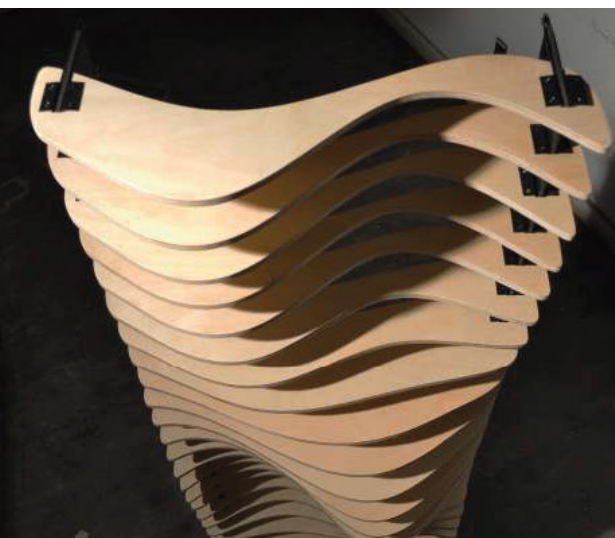
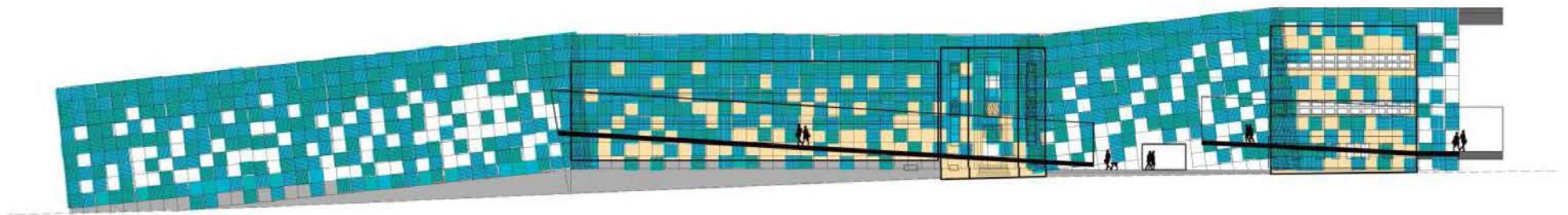
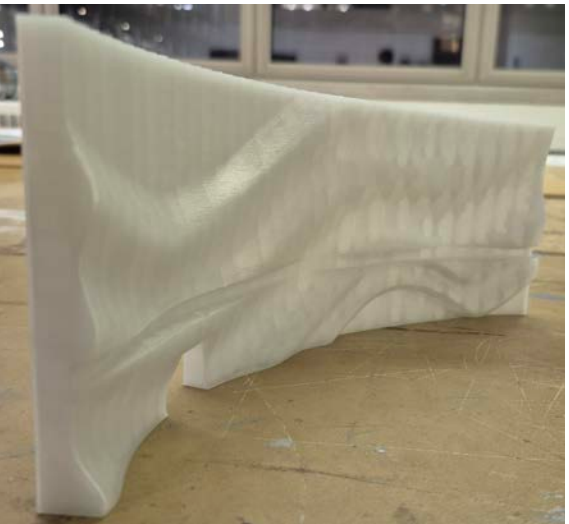
06/2019 UNAM Mexico City urban nature reserve - Rhino team

18 concept

04/2018 carpentry 3D lattice full scale - Rhino team

19 photos

20 design process



architecture for autism

strategy to destigmatize universal design

how can we densify the city in an inclusive way?

How to approach a marginalized neighborhood?

With transit-oriented development (TOD) plans and the implementation of a new tramway in LaSalle by 2050, it is necessary to plan for densifying this suburb, which has been marginalized by the city since Expo 67. This flagship project is built over an existing parking lot on Bergevin Street, two minutes from a new train station. The project revitalizes a fragile urban fabric marked by a gas line explosion that led to a fatal tragedy in 1965.

Moreover, the neighborhood is a food desert increasingly populated by migrants from visible minority communities. The strategy is carried out in three stages. First, the implementation of a new early childhood center (CPE). Next, the development of a new public square with commercial ground floors. Finally, the construction of new apartments for young adults.

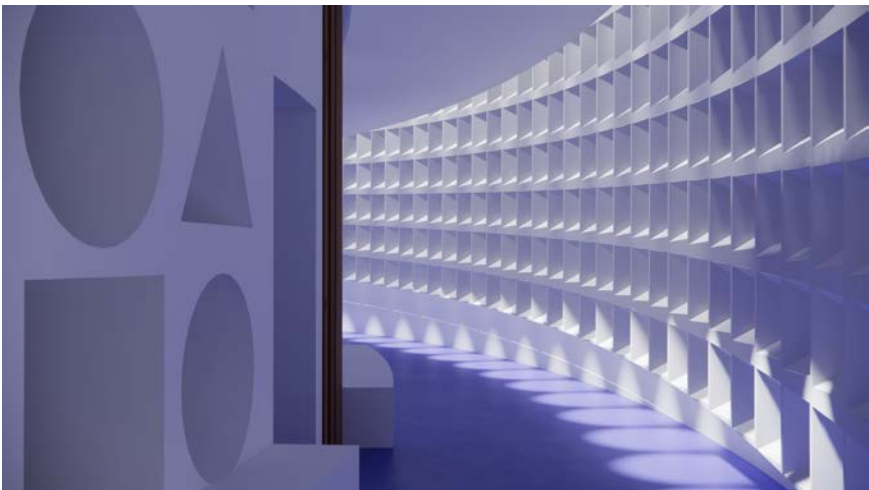
see a video tour and more images at akilgrunau.com



channel glass curtain



play spaces inside the belt



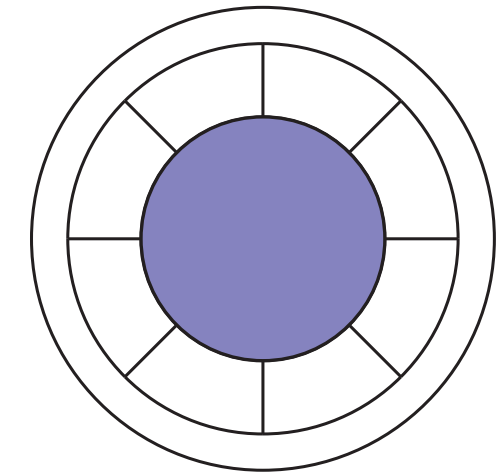
universal accessibility corridor belt

How should we think about territorial inclusivity?

A neighborhood populated by young families has tailored needs, particularly regarding safety, privacy, and physical accessibility.

Principle 1: Safety

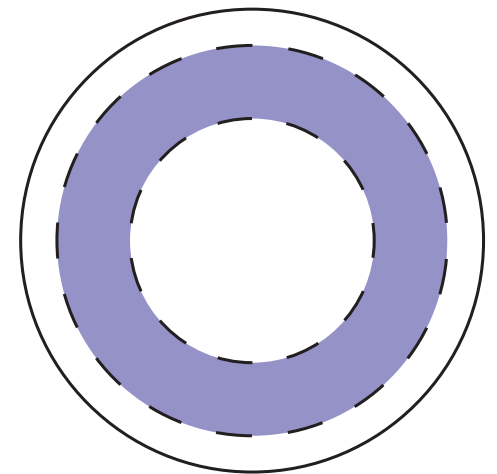
Early childhood centers (CPEs) are welcoming more and more children with behavioral disorders, ADHD, and on the autism spectrum (ASD). These centers receive additional funding from the Ministry of the Family (MF) to hire specialized educators. Since 2021, the MF requires one educator for 4 children aged 0 to 18 months, for 8 children aged 18 to 36 months, and for 10 children aged 3 to 5 years. It should be noted that many ASD cases are not recognized, especially among young girls. My project provides one educator for every three children, regardless of age and whether or not they have a diagnosis.



safety

Principle 2: Privacy

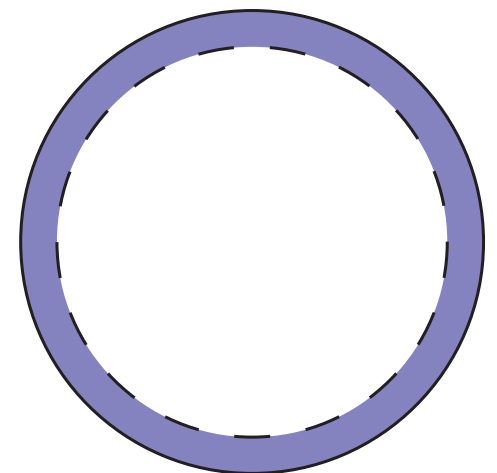
The International Convention on the Rights of the Child (CRC) established in 1989 that all children have the right to be well supported and loved. For a child to be well supported, their environment must be welcoming and inclusive; for a child to feel loved, the work of educators must be facilitated through spatial design. This flagship project proposes an intimate working environment for educators.



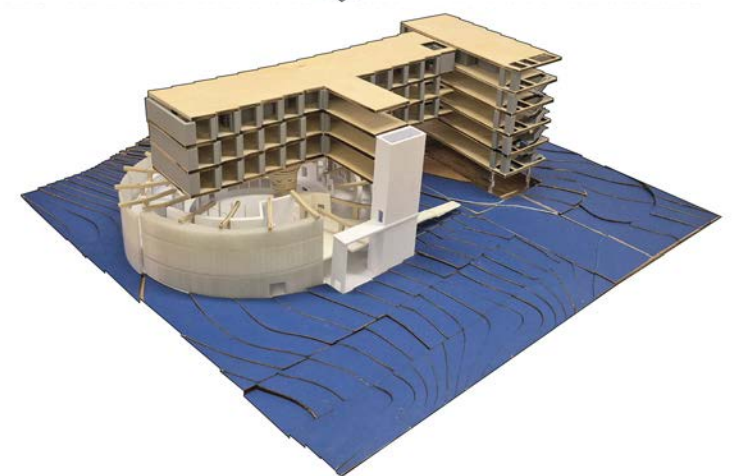
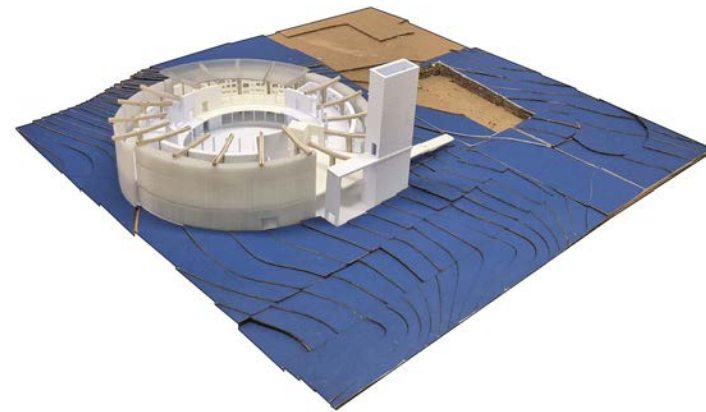
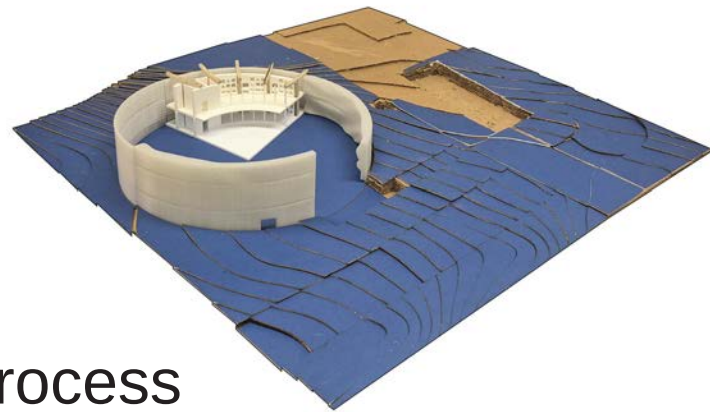
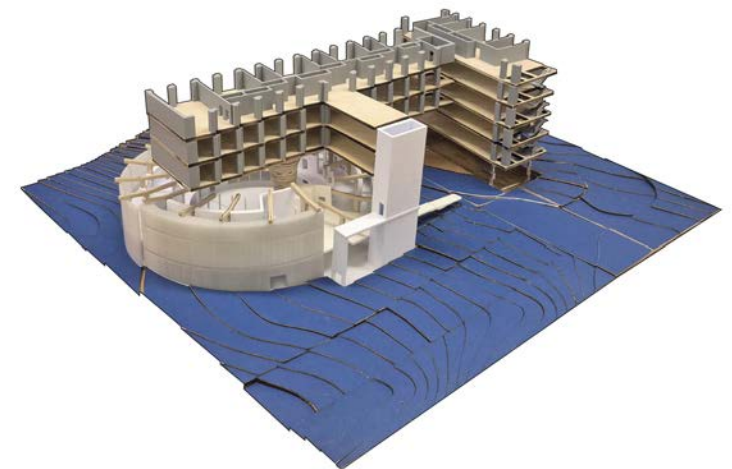
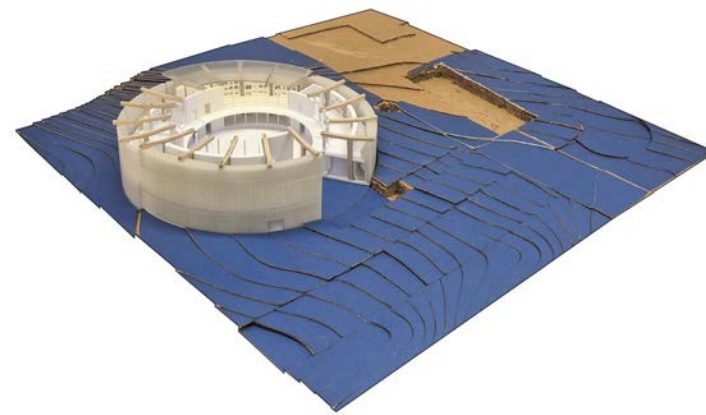
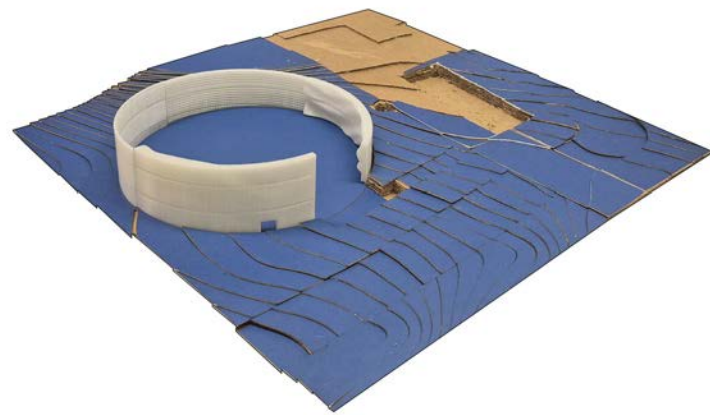
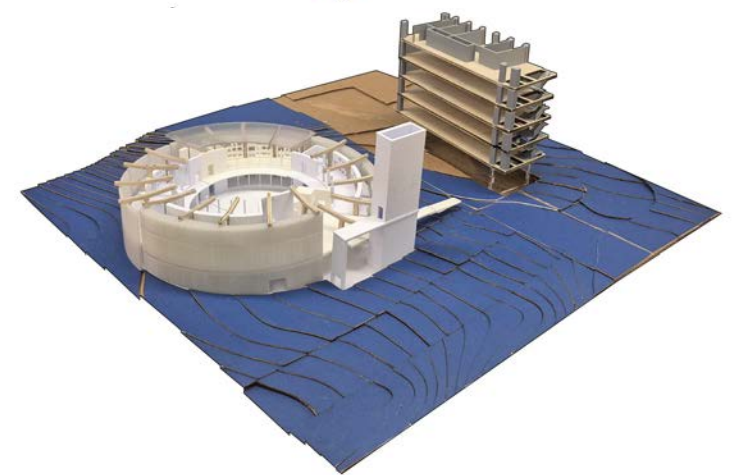
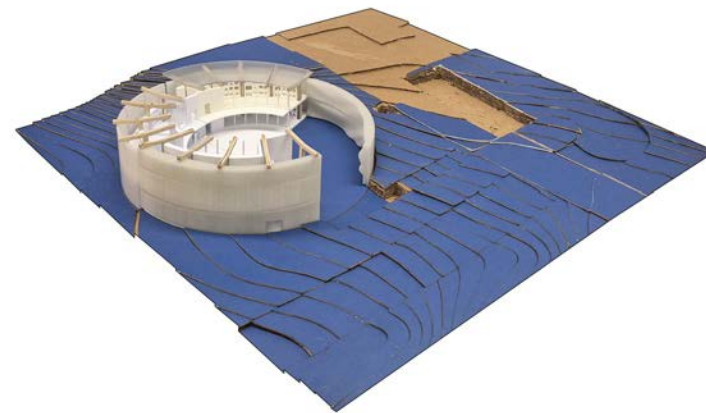
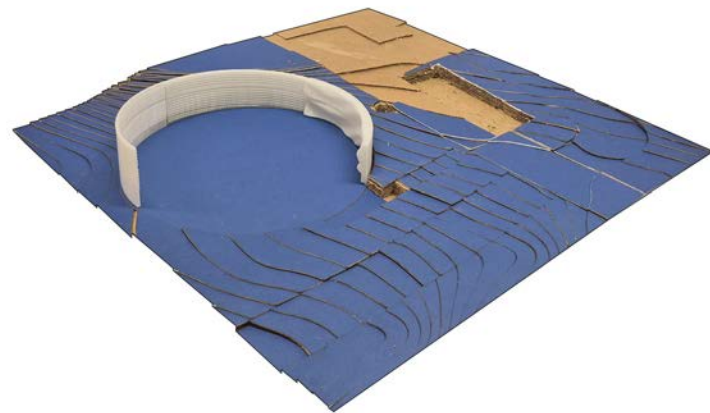
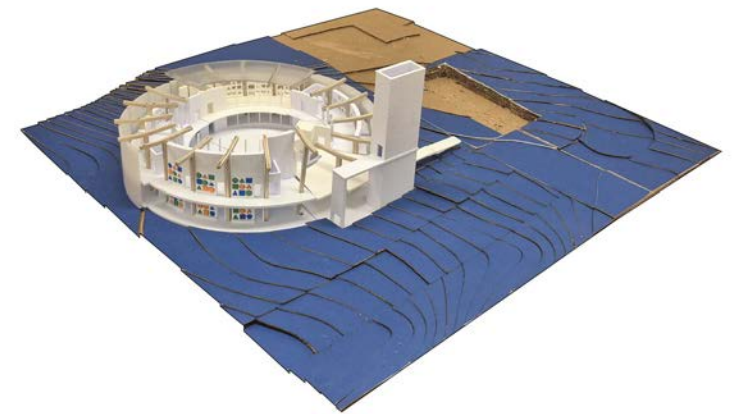
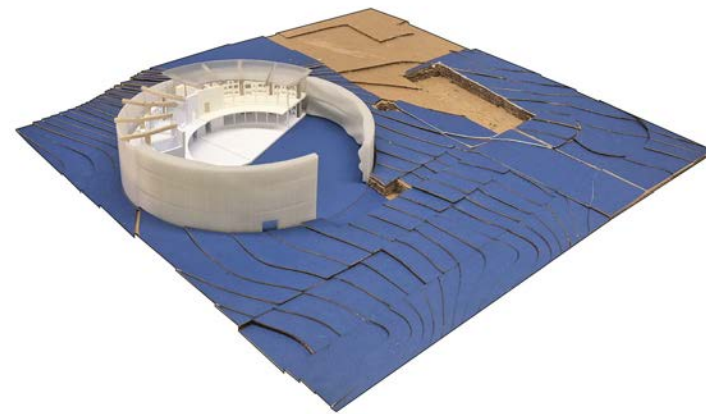
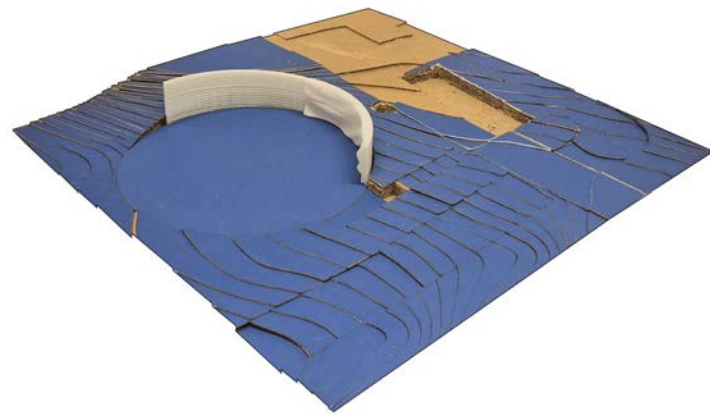
privacy

Principle 3: Physical Accessibility

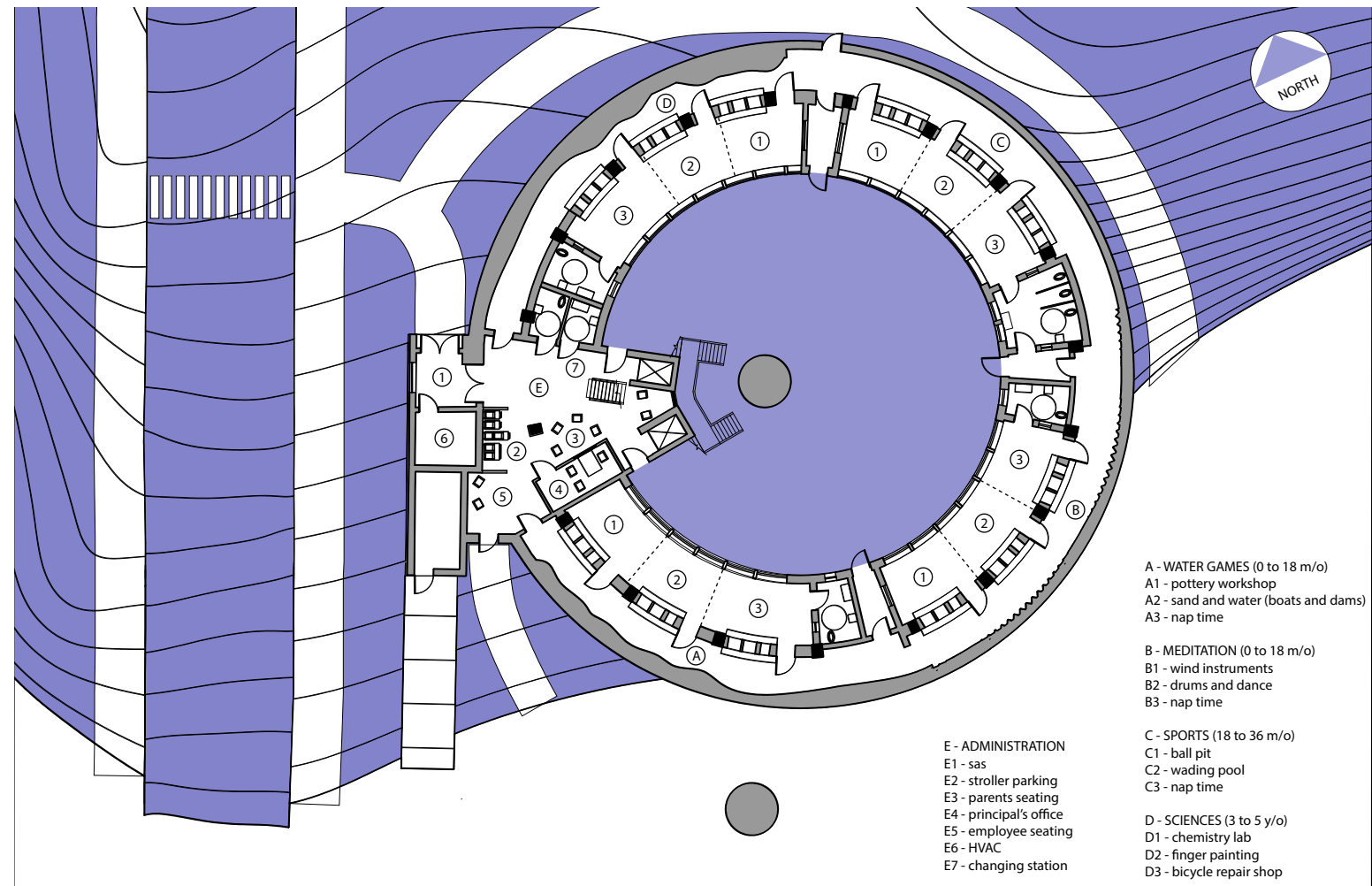
According to the CRC, every child has the right to freedom of participation. To achieve this, an inclusive and accessible environment must be designed for all children. This CPE is a flagship project for inclusion and accessibility that adapts to each child's conditions to strengthen their sense of identity and support a degree of autonomy. Providing sensorial indications to navigate space-time is part of the solution.



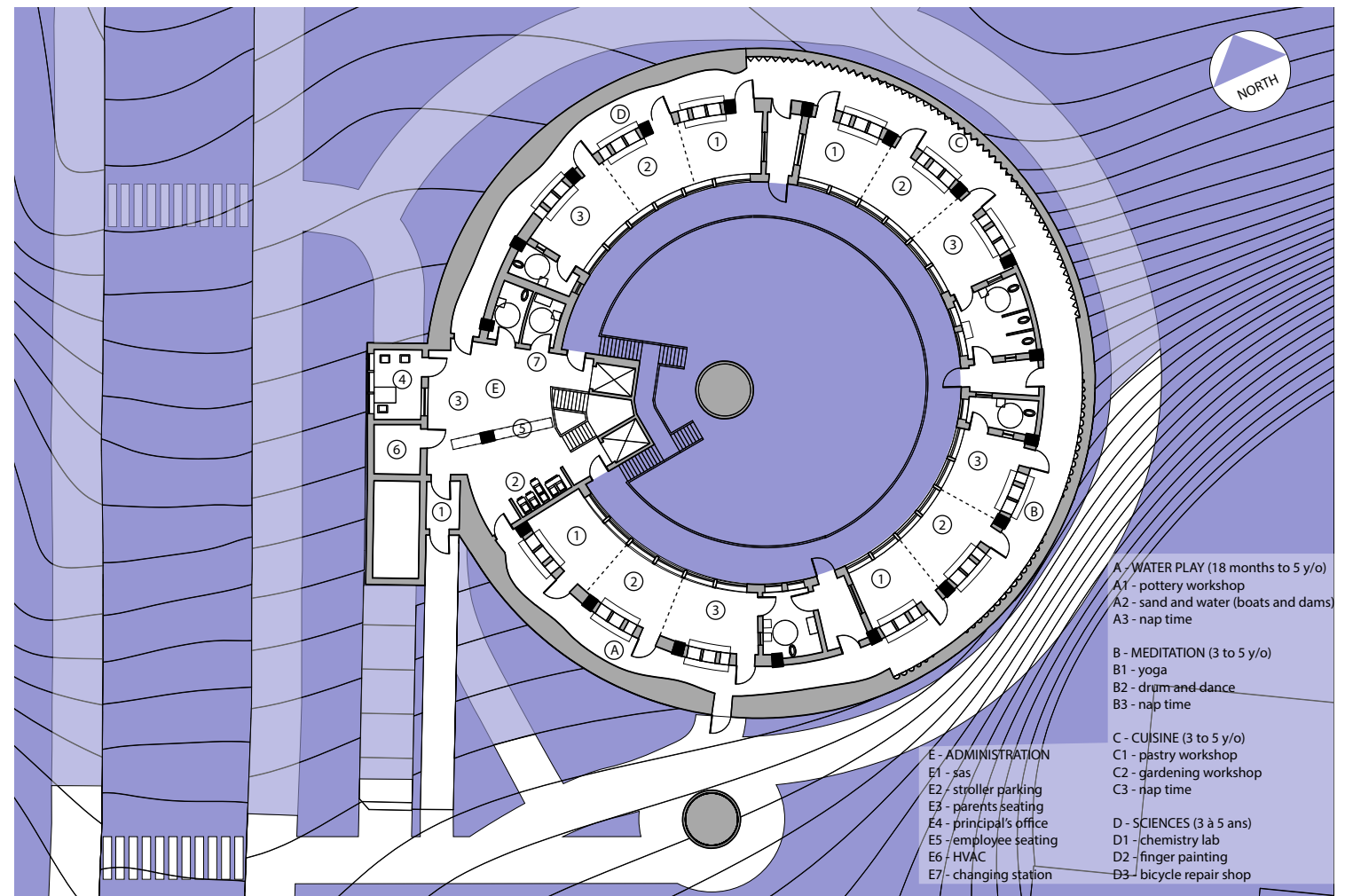
universal accessibility



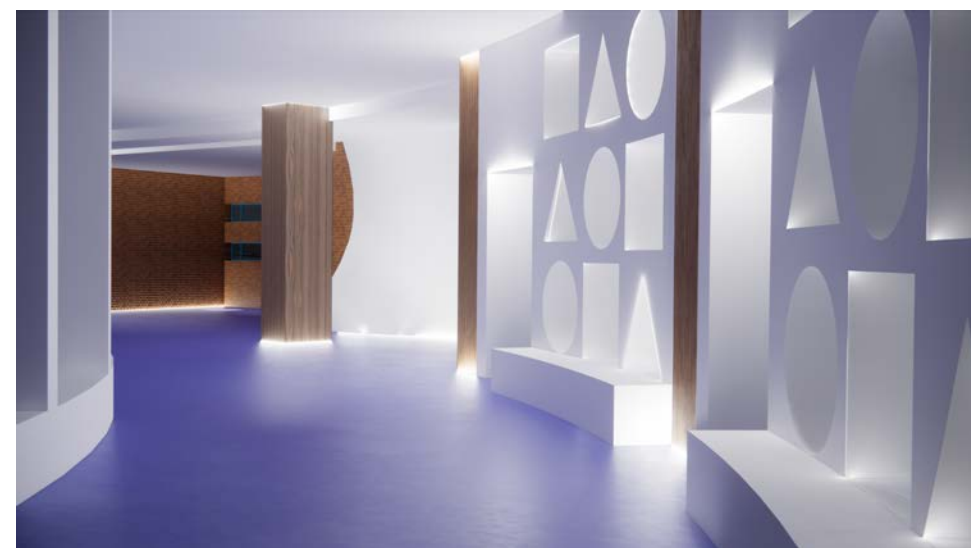
construction process



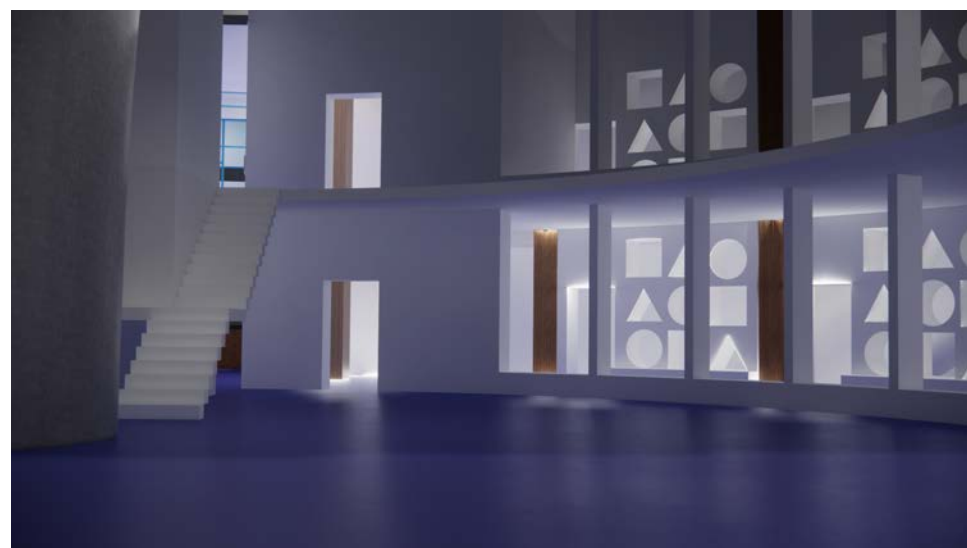
Level 1



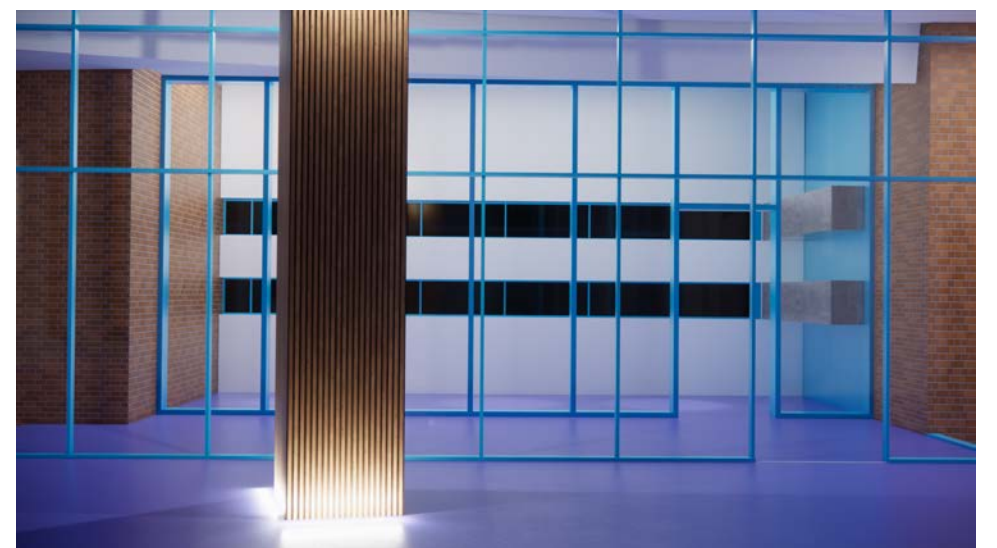
Level 2



indoor play area and changing stations

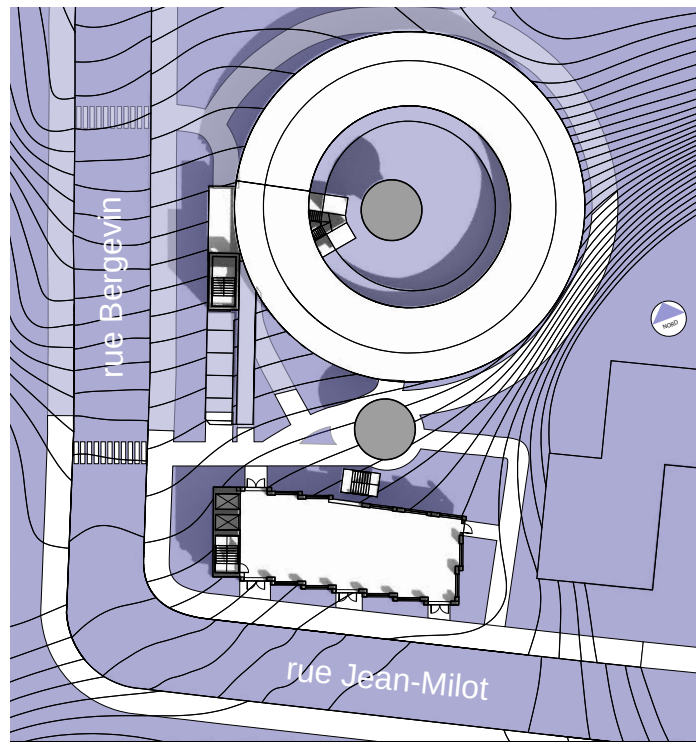


outdoor play area

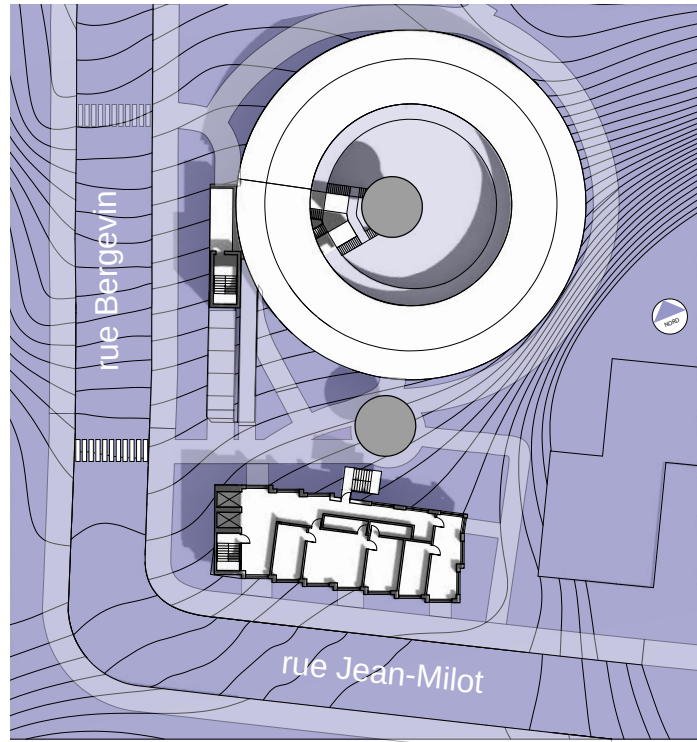


E1 - entrance - Level 2

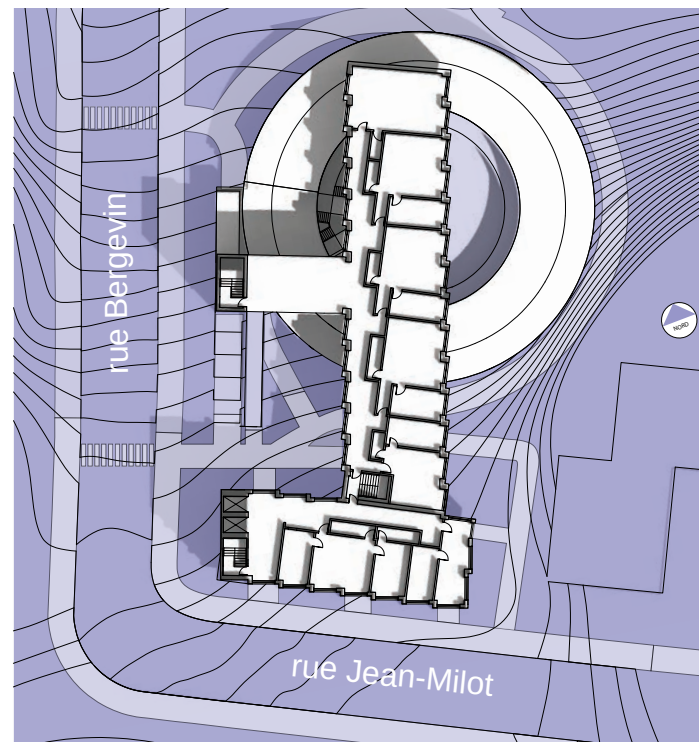
early childhood learning center



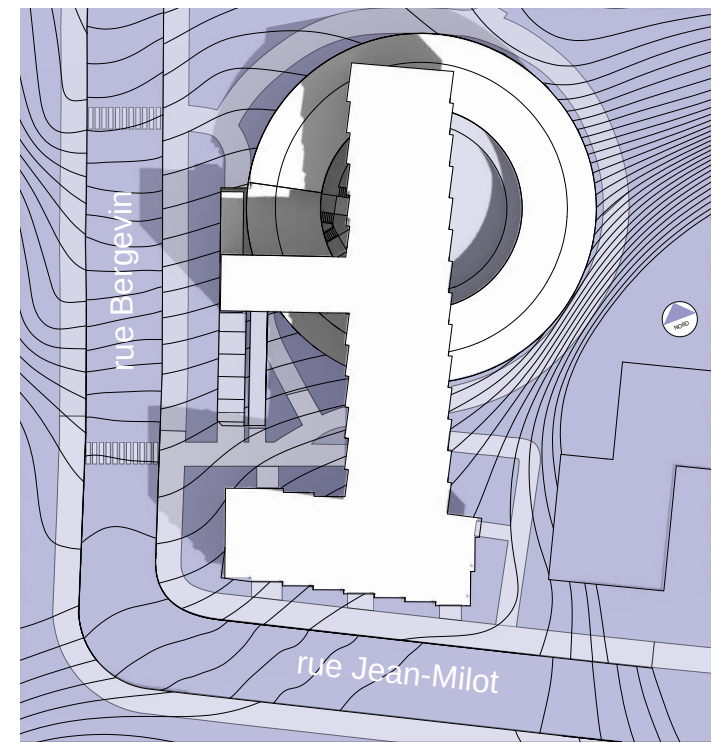
Level 3 - commercial ground level



Levels 4, 5 - apartments



Levels 6, 7, 8 - apartments and common area

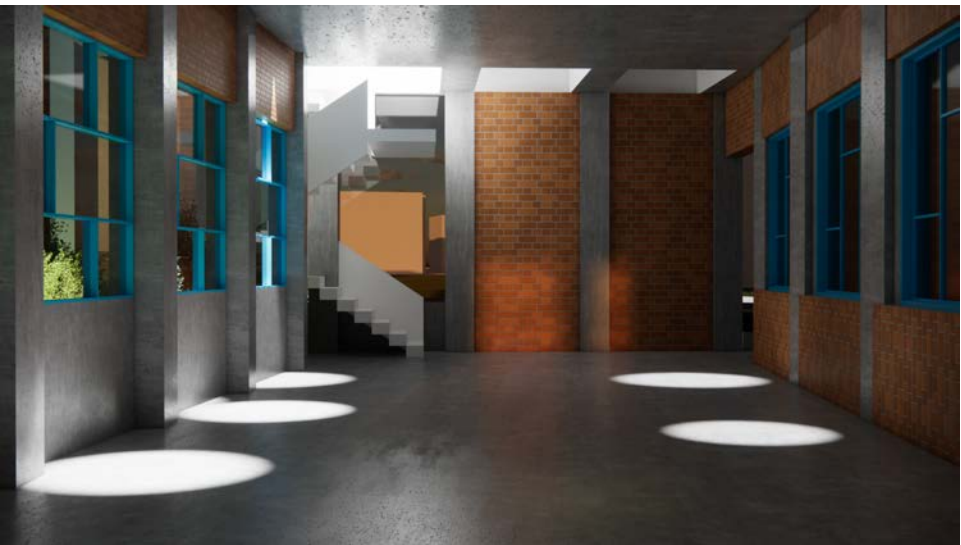
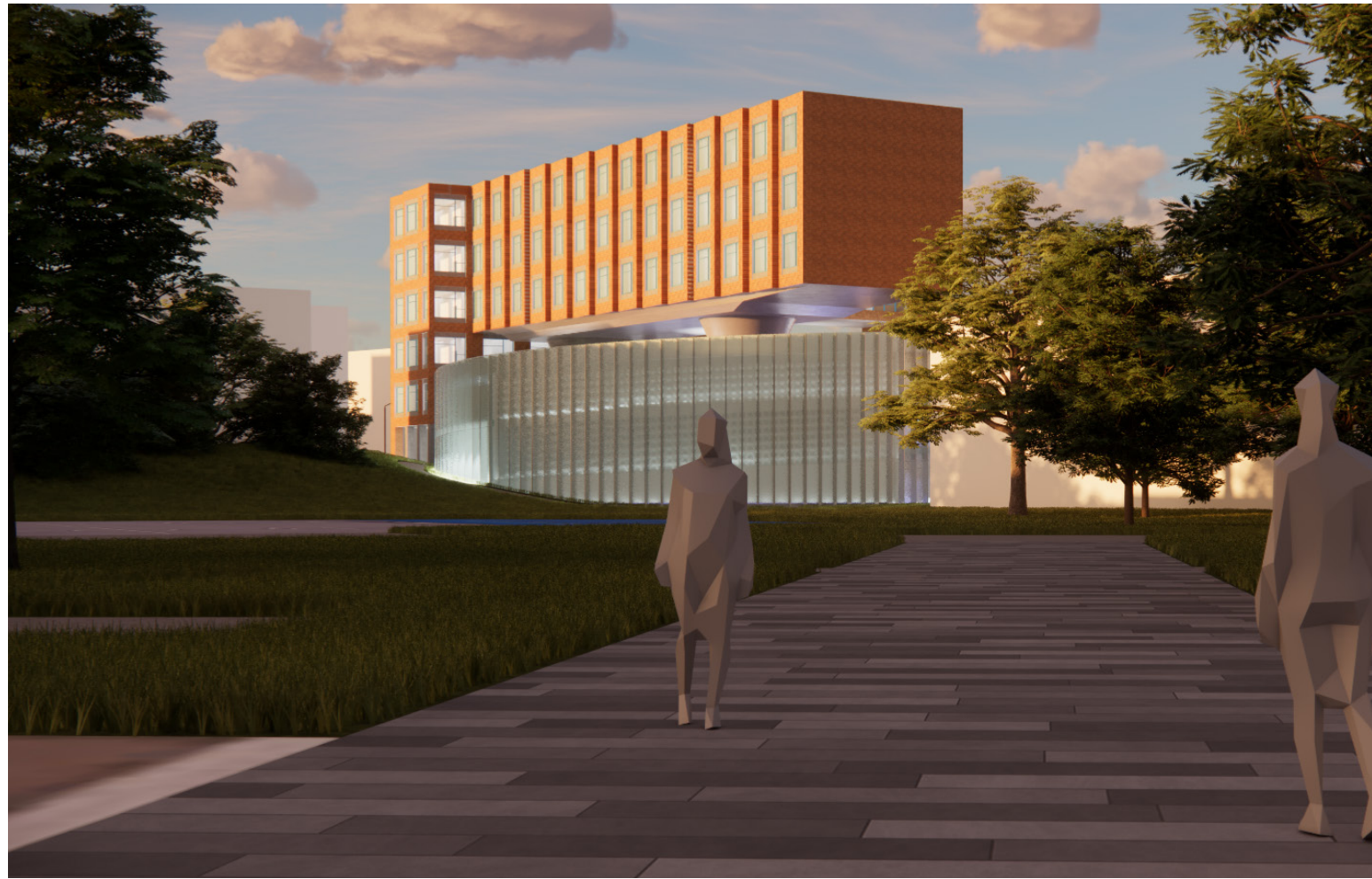


Roof





The glass curtain provides a new ornament to the parc and a neighbourhood symbol of hope

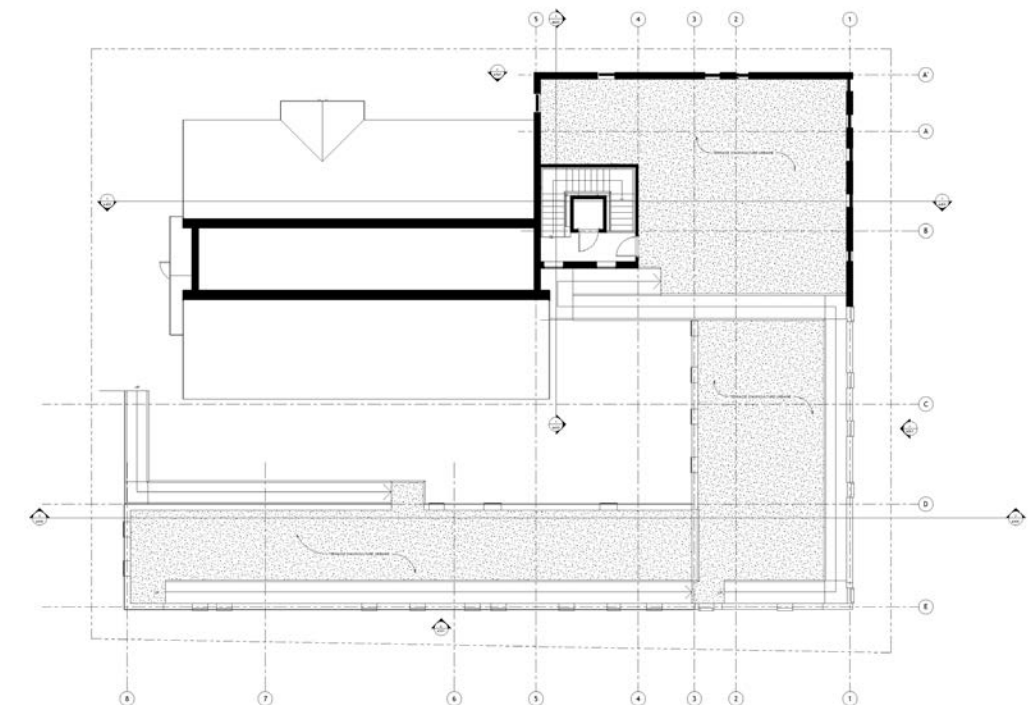
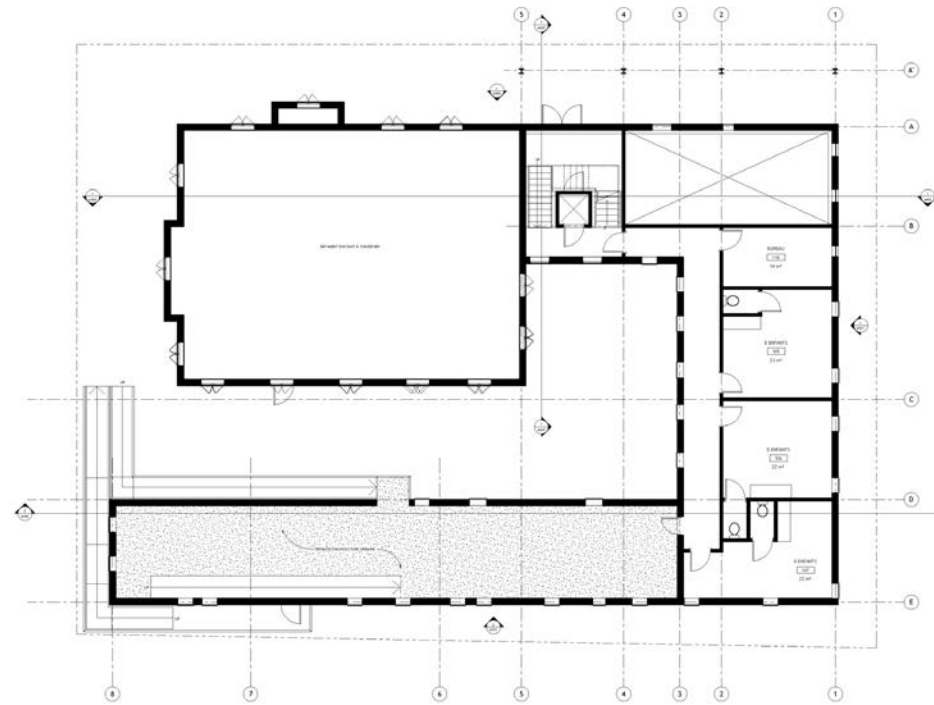
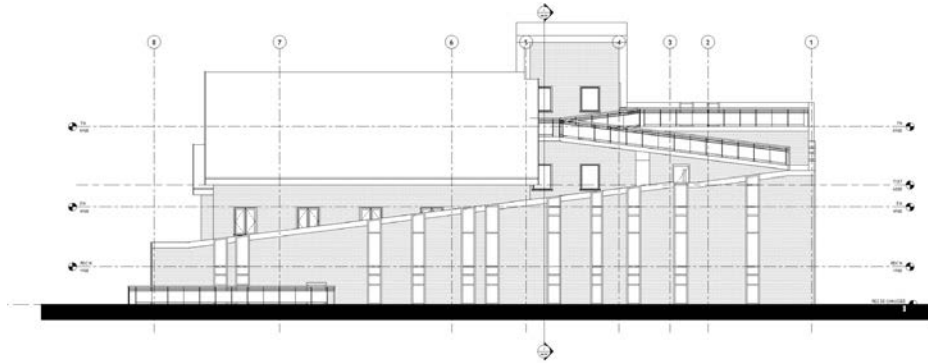
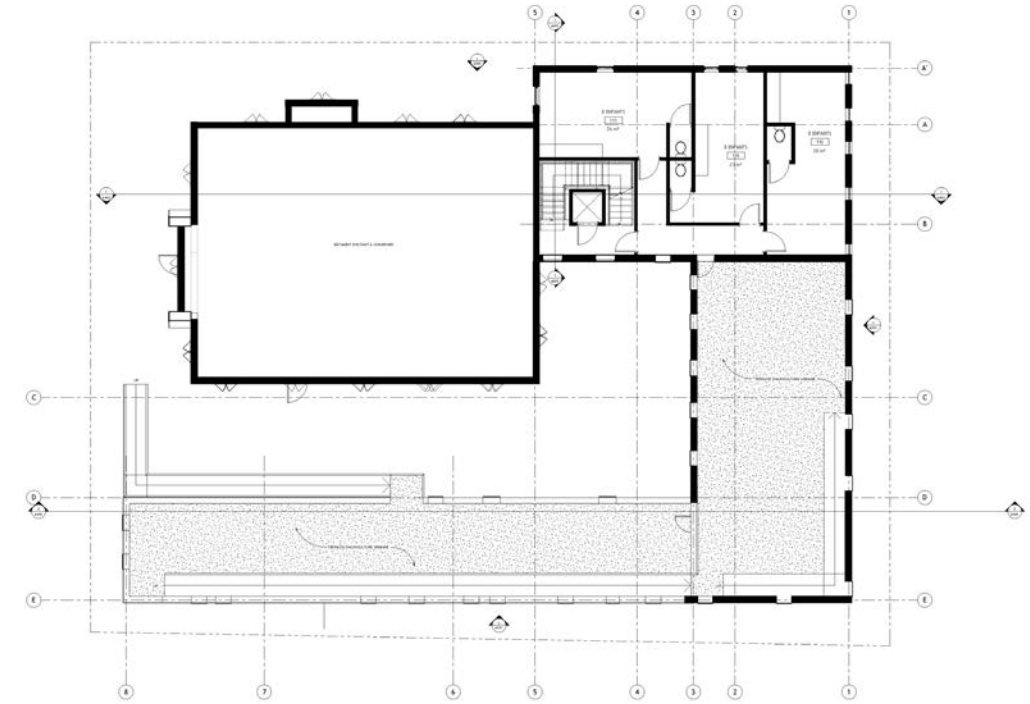
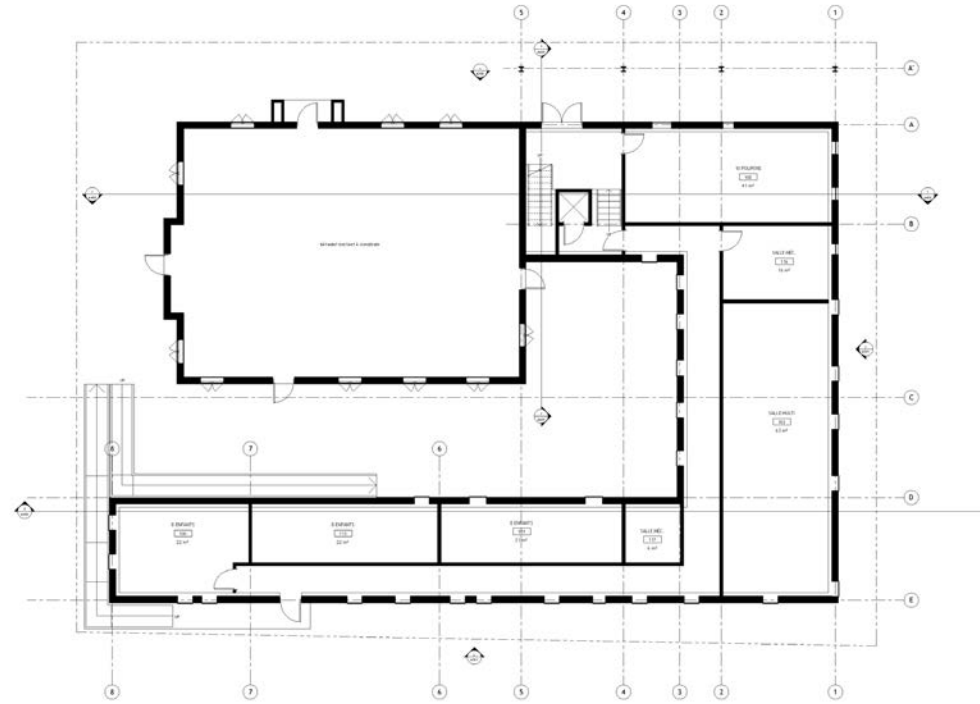
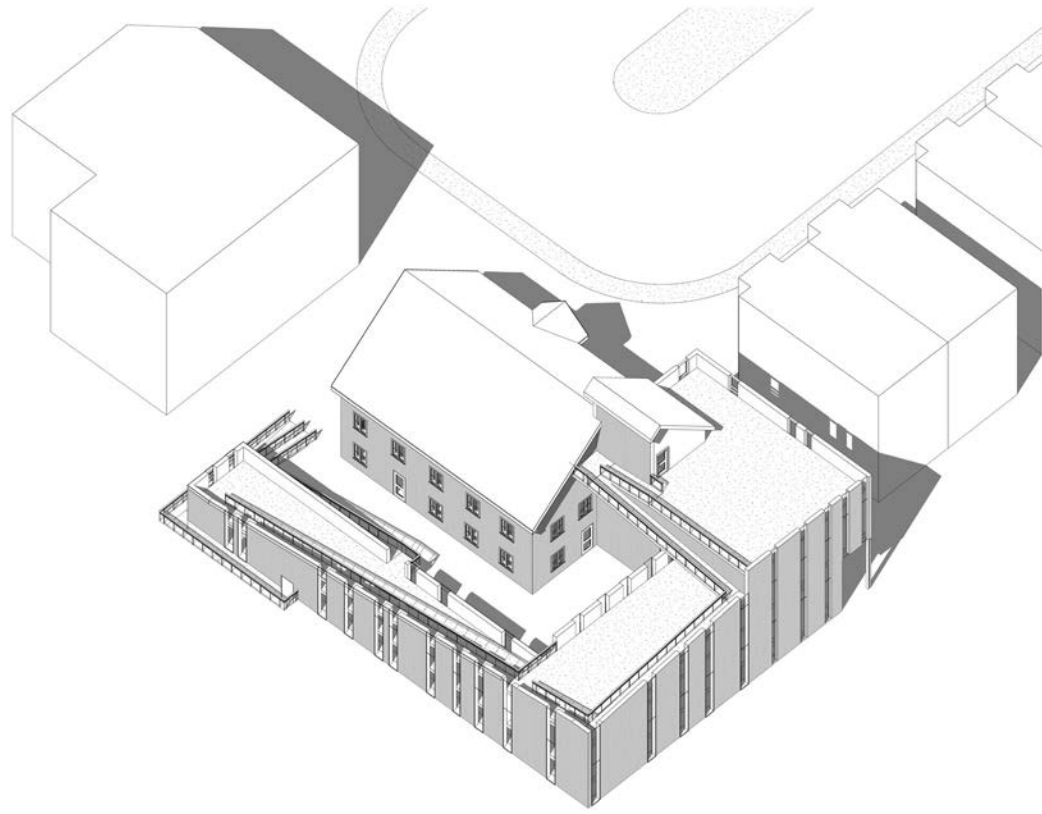


Café space facing rue Jean-Milot

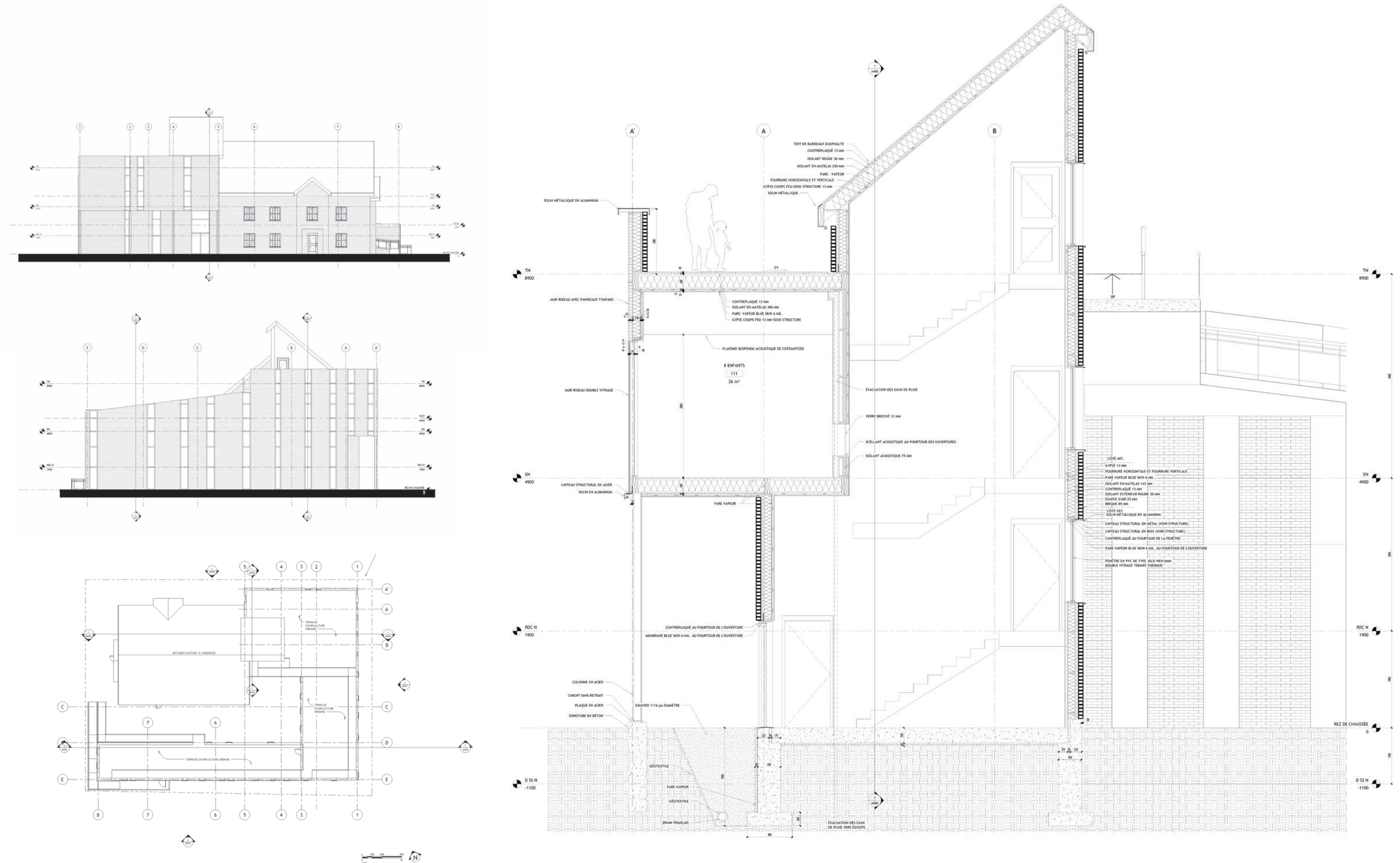


The garden separates the café from the sidewalk space





CPE Early childhood learning center - Revit project



CPE Early childhood learning center - Revit project



Existing apartments

Crosswalk

Existing building

New extension

Existing bungalow

To better blend the building extension with the existing we proposed the following solutions to the urban planning committee of montreal - hochelaga-maissonneuve

A copper treehouse pertrubes from the facade to highlight the new entrance and separate the old brick from the new colour scheme of mango and cream bricks.

A mansard roof was chosen to provide bangs to the new facade while reminding the viewer that the old facade bangs are simply too short. Sliding grey metal siding was chosen with house shaped pertrusions at each window.

The new mango cream colour scheme provides the school with a new identity while still working with the original logo and providing hope for change in this grey neighbourhood.

By city standards every facade should be 80% masonry excluding glazing yet our 60% masonry proposition was accepted by the council.

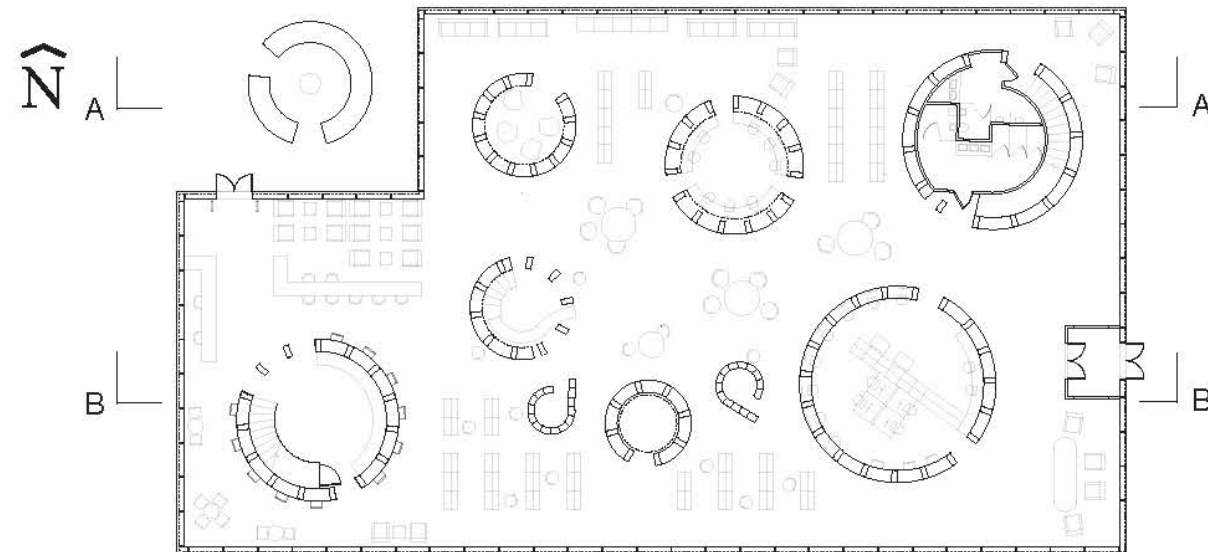
3D model, street section and perspective produced by Akil Grunau as an employee of Tracé Libre Inc.



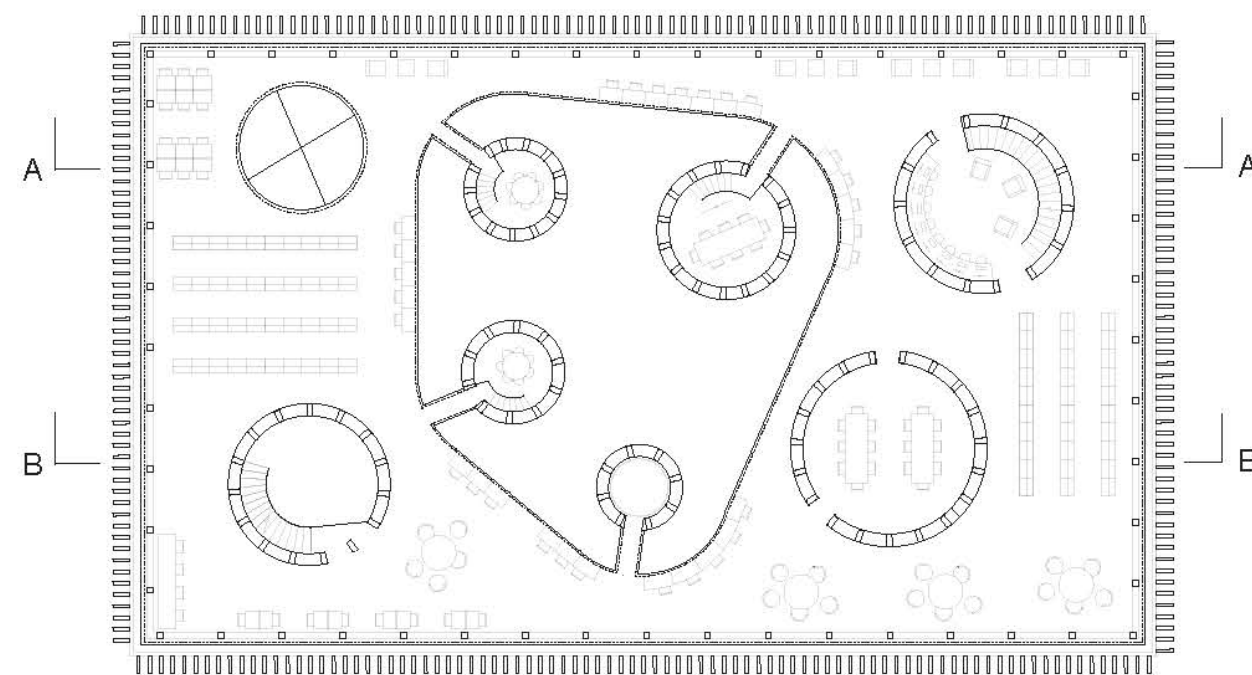
Early childhood education center extension (public project)
Julien Valade arch, Clémentine Dambricourt arch, Akil Grunau tech
copyright Tracé Libre Inc. 2023



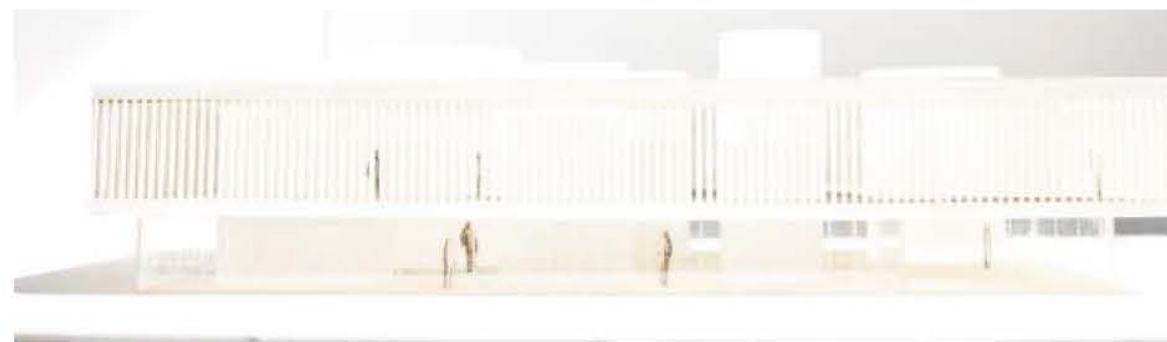
Plateau public library



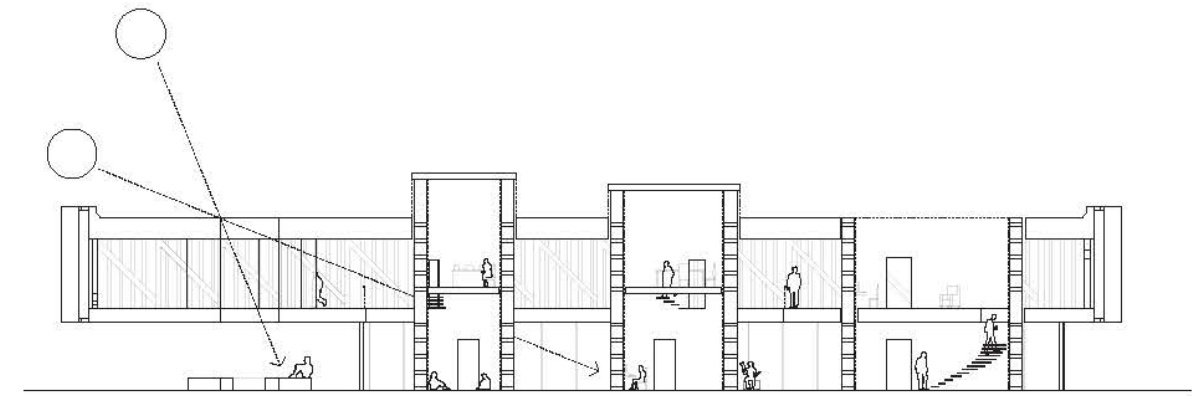
ground level plan



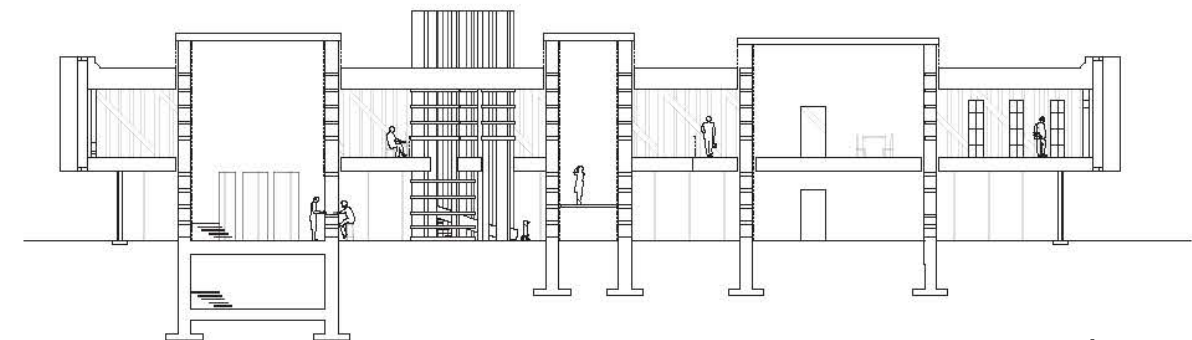
second level plan



card stock model



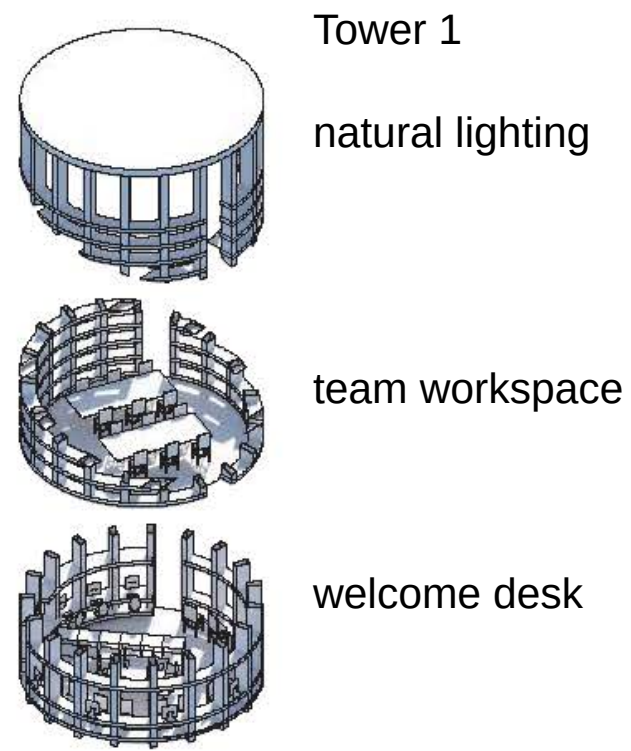
section A



section B

The book towers vary in height, width and program. Certain towers have stairs, others have a lookout, a play space, a quiet space, a work space or a social space. These hollow columns are the gears that shift the buildings velocity throwing the chainlink of visitors from one to another. The main entrance is placed near the louder towers while the exit is adjacent the cafe and outdoor pow wow. The library is open from 10 AM to 10 PM.



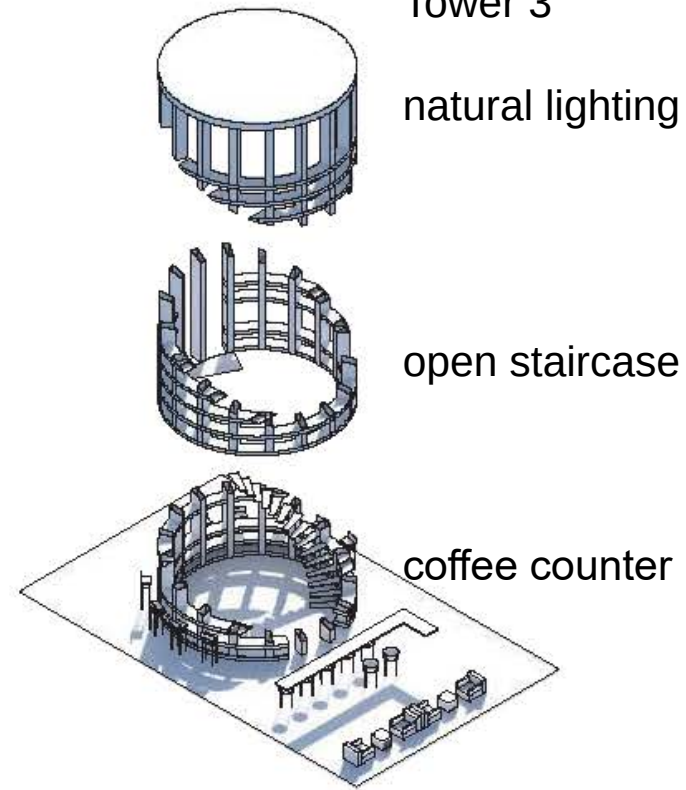


Tower 1

natural lighting

team workspace

welcome desk

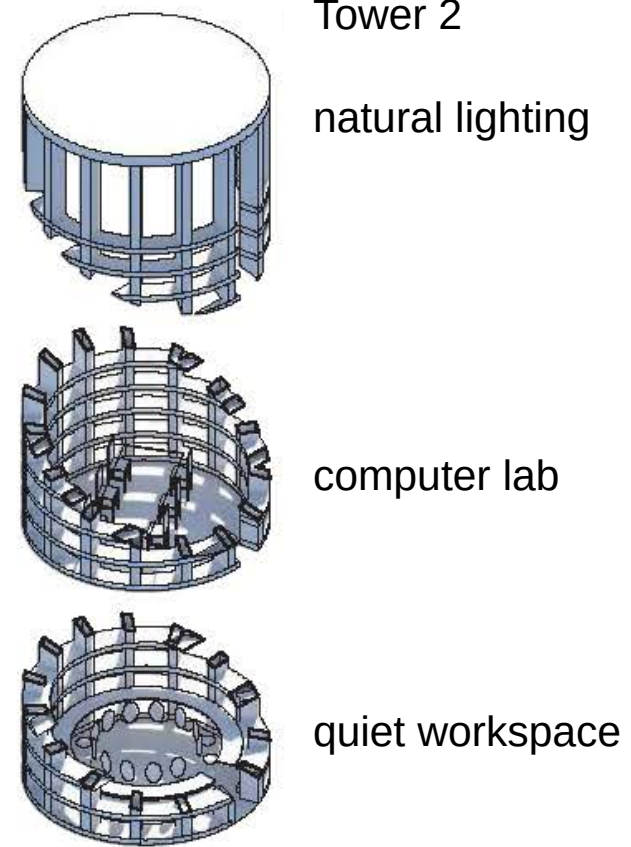


Tower 3

natural lighting

open staircase

coffee counter

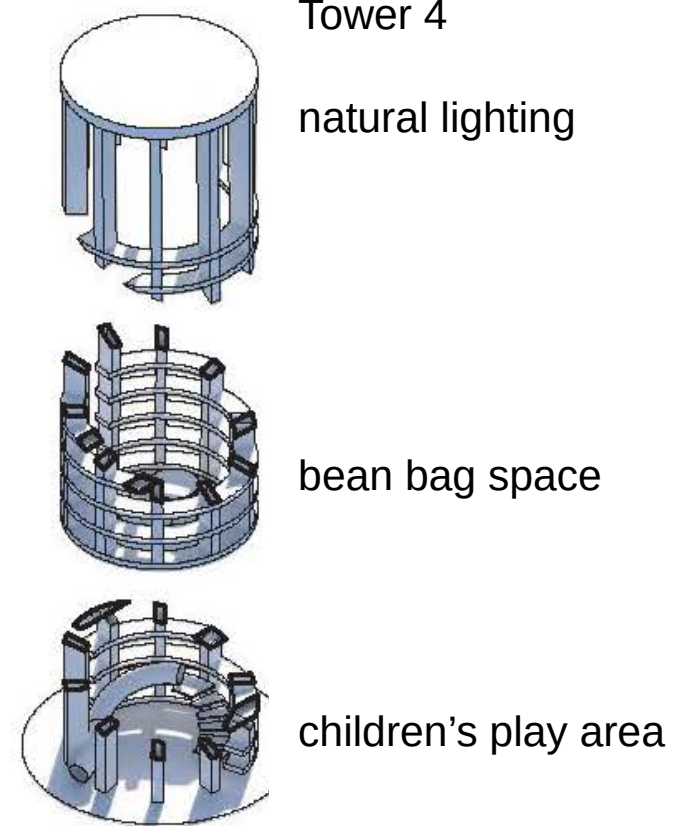


Tower 2

natural lighting

computer lab

quiet workspace

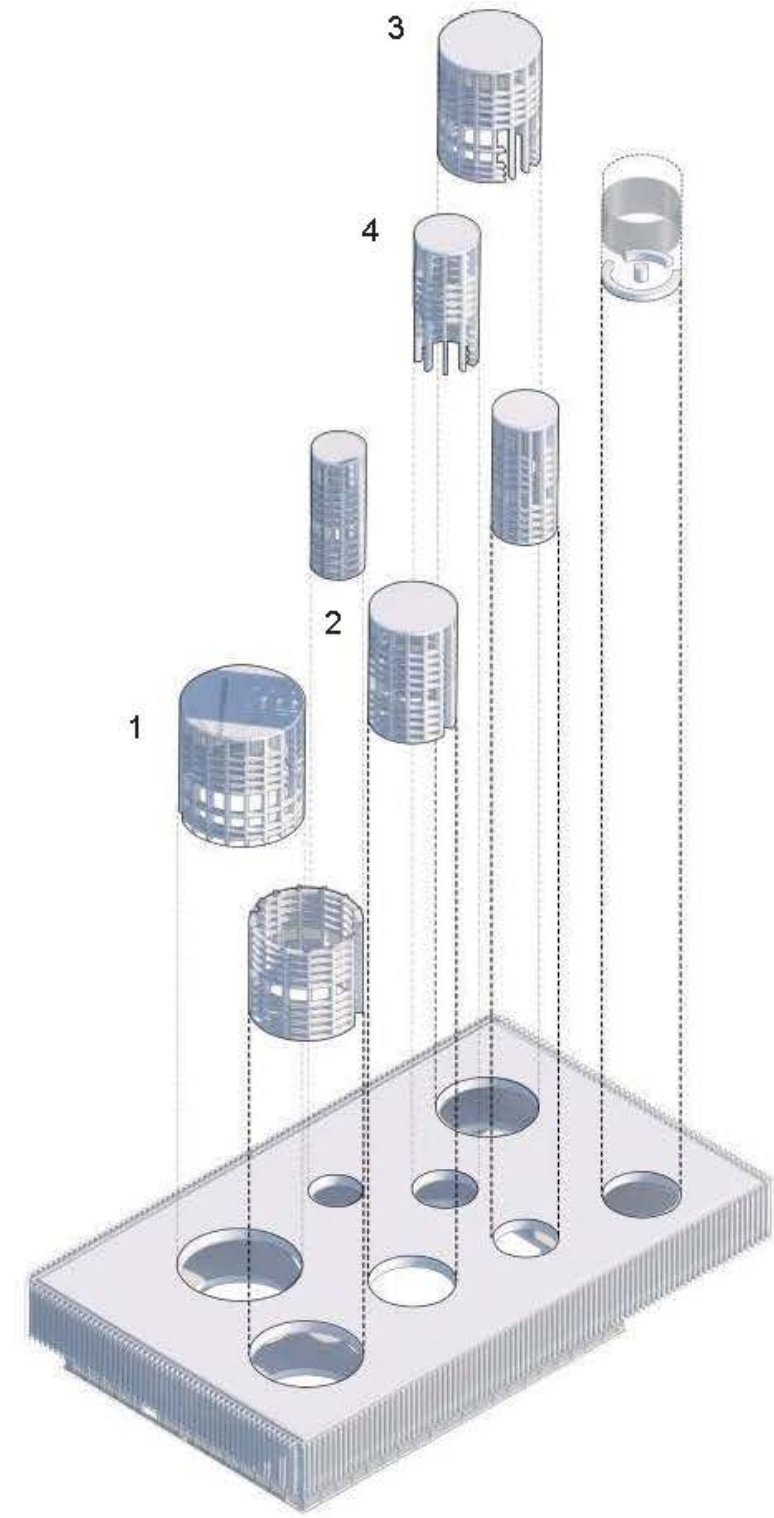


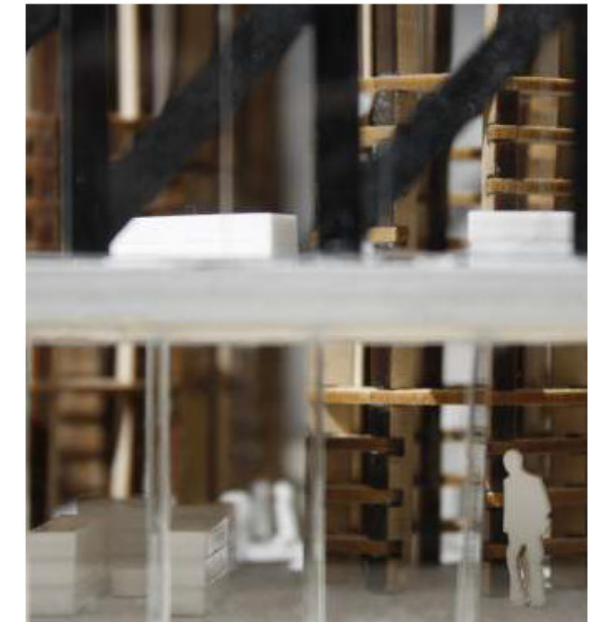
Tower 4

natural lighting

bean bag space

children's play area

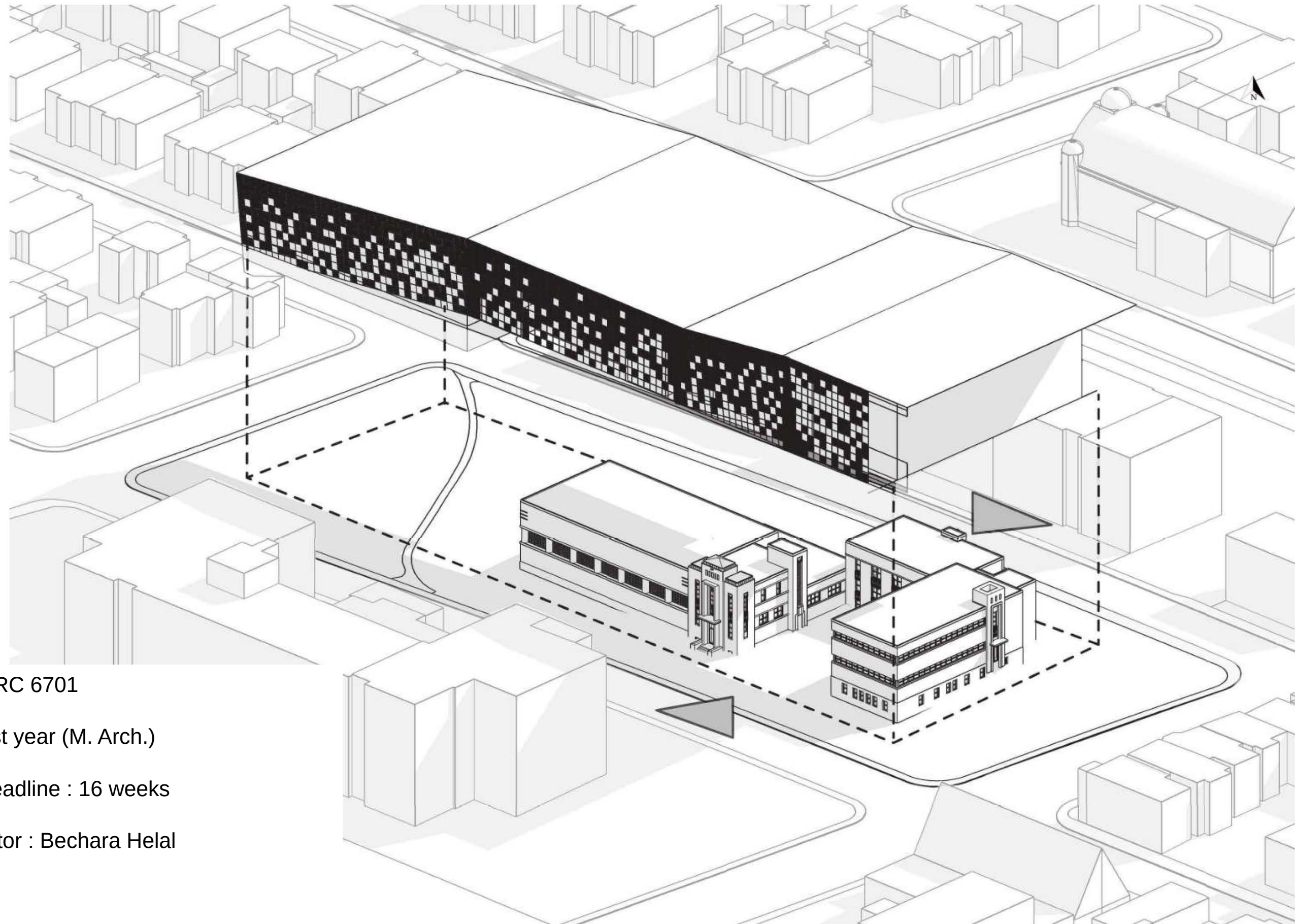




The children's play area is the heart of the building and the most versatile space with light furniture.



Pow wow space adjacent to café provides a bridge for nighttime community gatherings towards the exit.



ARC 6701

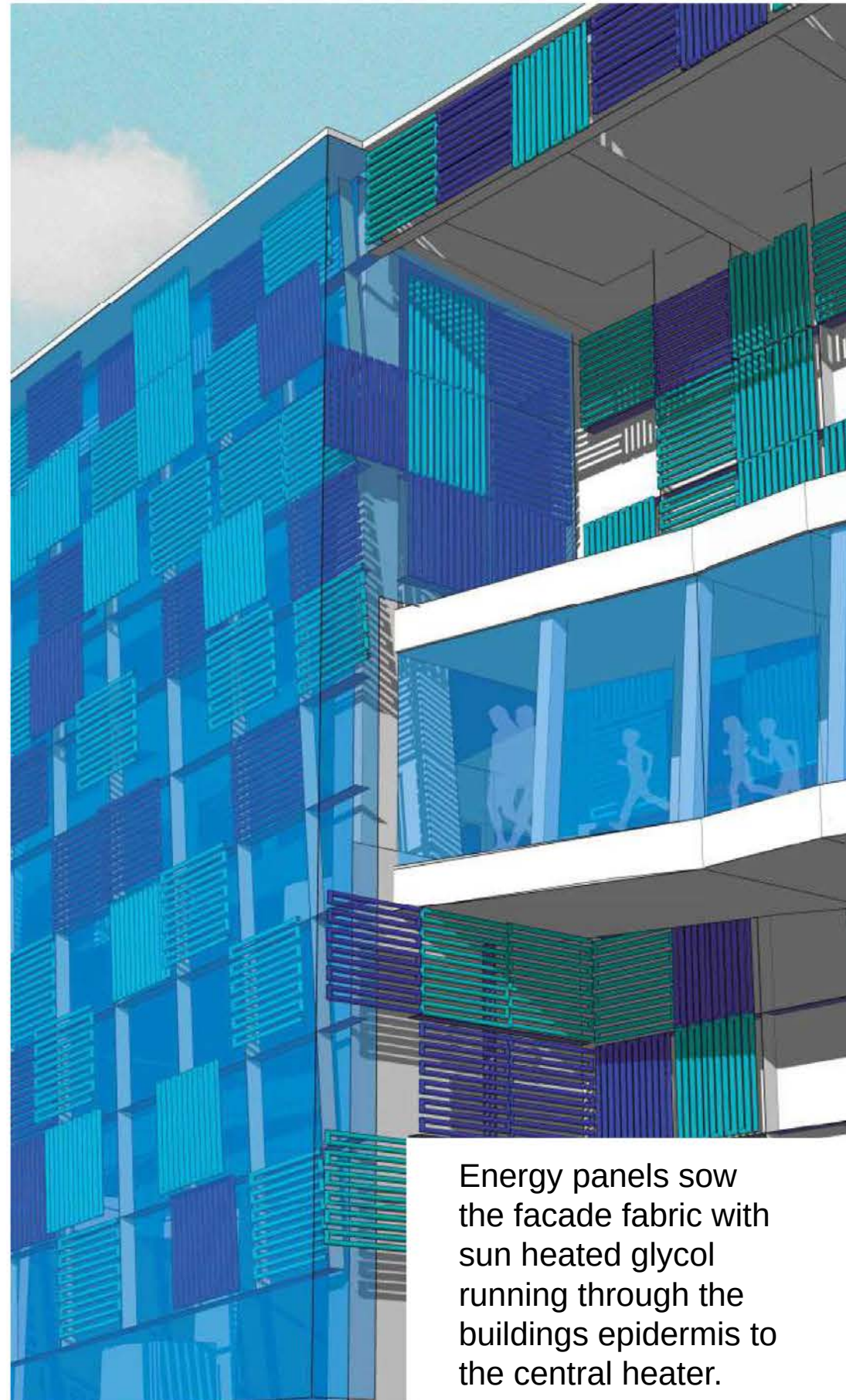
1st year (M. Arch.)

deadline : 16 weeks

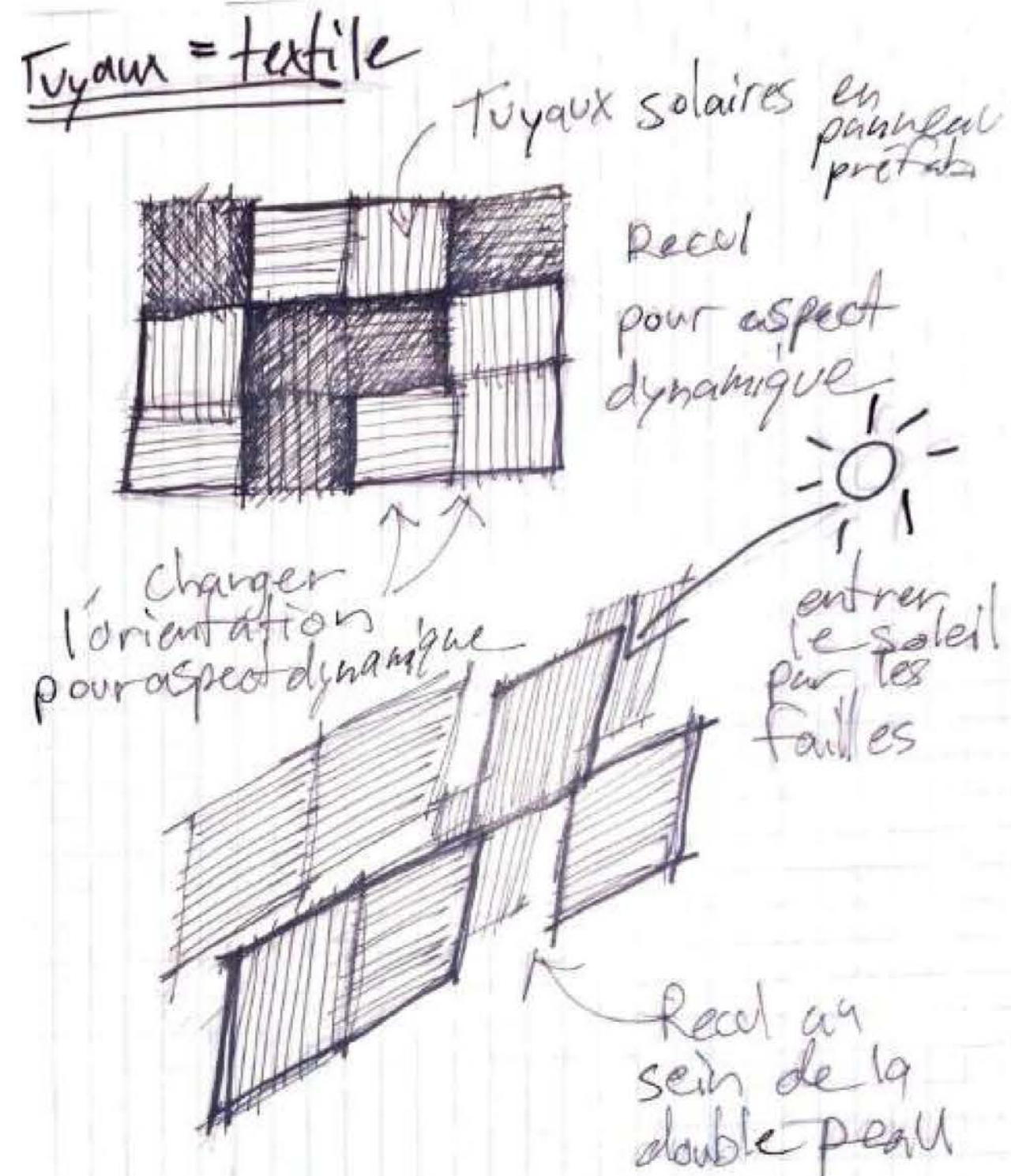
tutor : Bechara Helal

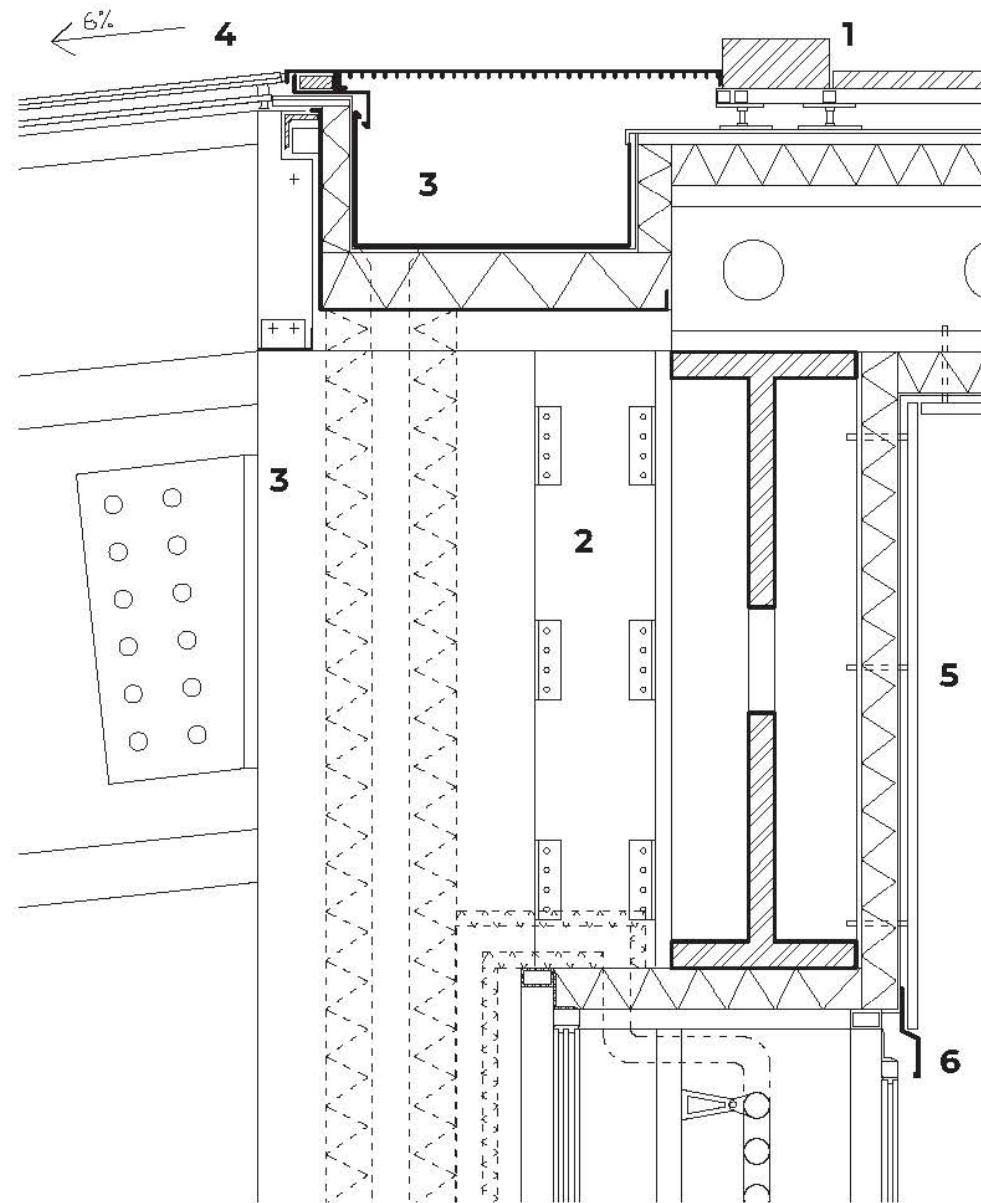
context and concept illustration

Rosemont community center *Université de Montréal*
winter 2019



Energy panels sow the facade fabric with sun heated glycol running through the buildings epidermis to the central heater.



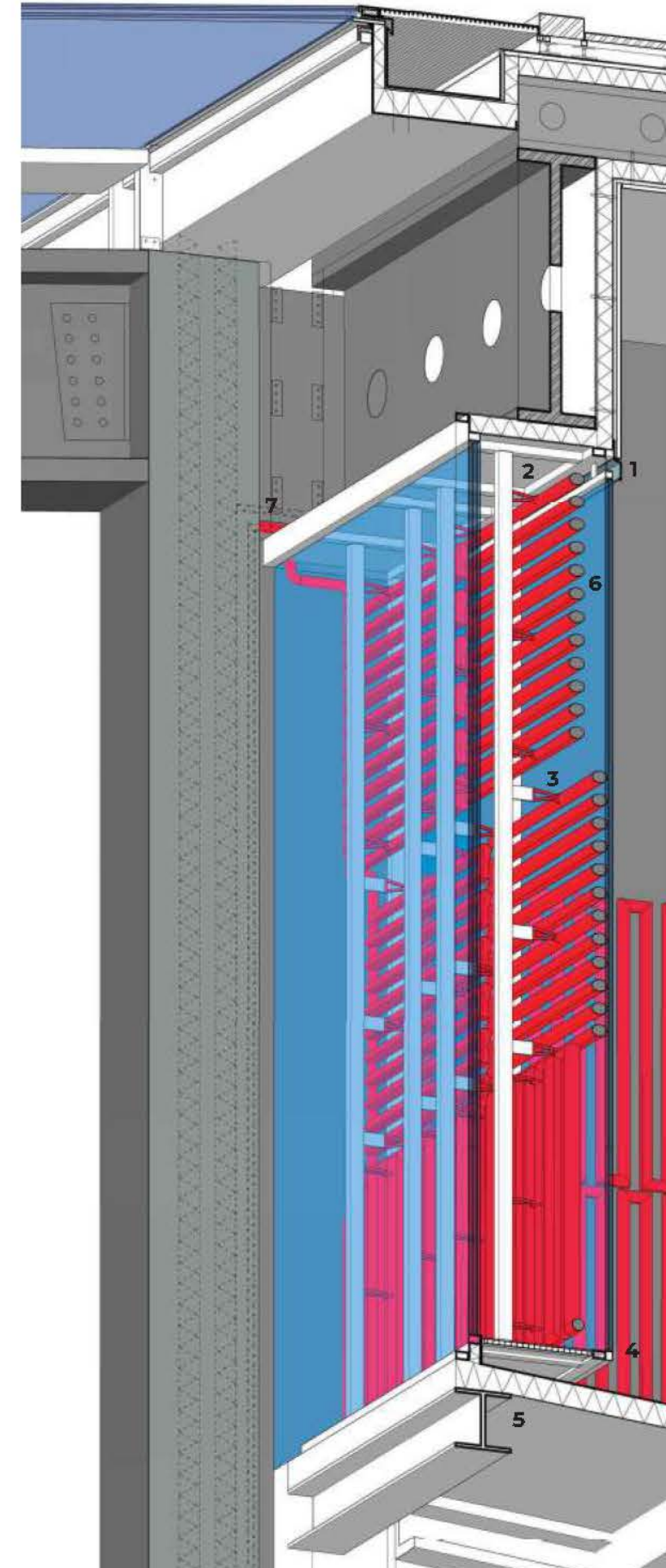


construction detail section

- 1- roof concrete panel, steel frame, grace ice and water shield, OSB, foam board insulation (80 mm), vapor barrier, steel beam
- 2- steel column for pool structure and steel beam for separate structure bolted together through a shock absorber joint
- 3- rubber sheeting, aluminum gutter (525 mm wide), sprayed fibreglass insulation (100 mm), steel flashing
- 4- triple glazing fixed with steel connectors
- 5- prefab concrete panels, windbreaker, fibreglass sprayed insulation (70 mm), vapor barrier, OSB, steel beam
- 6- outer double glazing and inner triple glazing heat window system with a 600 mm air gap

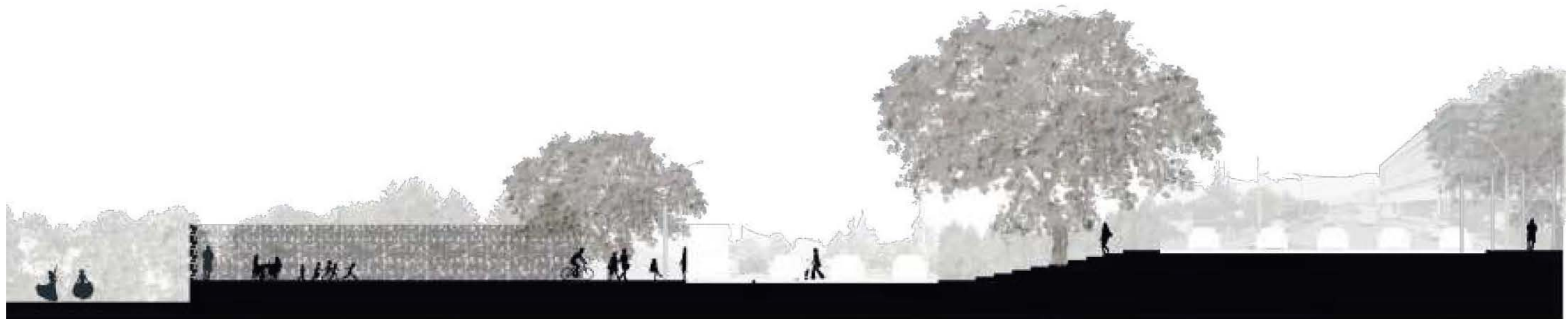
axonometry detail

- 1- steel flashing opens the hot air exhaust
- 2- steel frame
- 3- energy panels attached to the steel frame through a movement joint
- 4- cool air intake
- 5- steel beam
- 6- copper tubes patchwork energy panels connected through a glycol fluid network for central heating from sunlight
- 7- copper tubes network insulated throughout the building interior



construction detail section and axonometry

Rosemont community center *Université de Montréal*
winter 2019



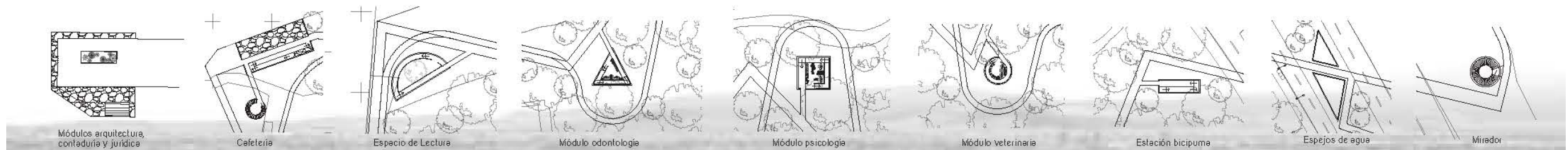
university (UNAM)

semi-public space

public space

university residence

cross-section Avenida del Imán Mexico City



new academic and transportation nodes across the nature reserve for student activities and the preservation of shared culture and knowledge



CONCLUSIÓN

Se crea una vinculación entre CU y su entorno, puesto que se brinda un espacio de carácter público en donde encontrarán diversas actividades. El espacio público funciona como base estructural de la ciudad, donde se expresa la memoria colectiva, es así como se hace ciudad con Ciudad Universitaria.

The nature reserve that connects the university to the city

ARC 2011

2nd year (B. Arch.)

deadline : 3 weeks

tutor : Morgan Carter

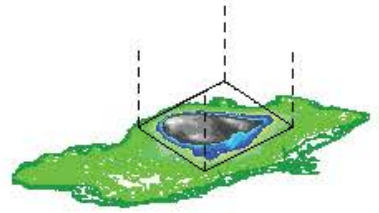
team project with
Karl Lessard
Rosine Tétrault
Charles-Étienne Bilodeau

solar lattice
mock-up

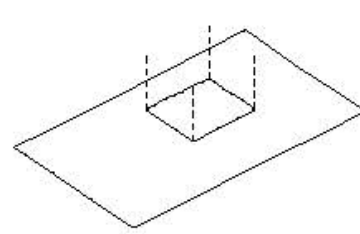
photo of mock-up



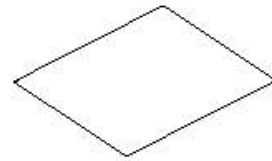
1 - topography



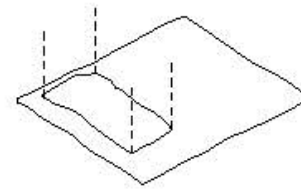
2 - extract



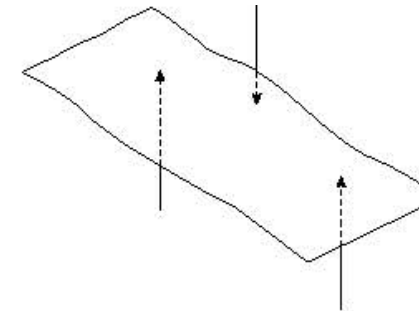
3 - rough surface



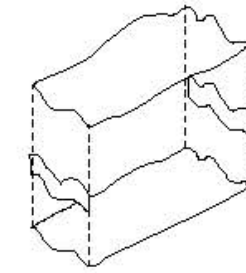
4 - fragment



5 - simplification



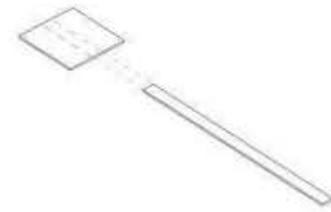
6 - slice



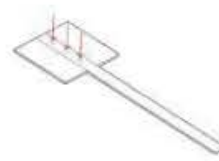
1 - cut



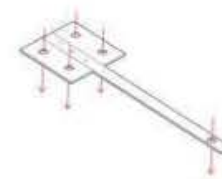
2 - mark



3 - weld



4 - pierce



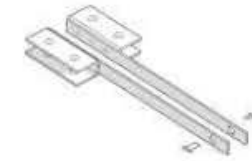
5 - fold



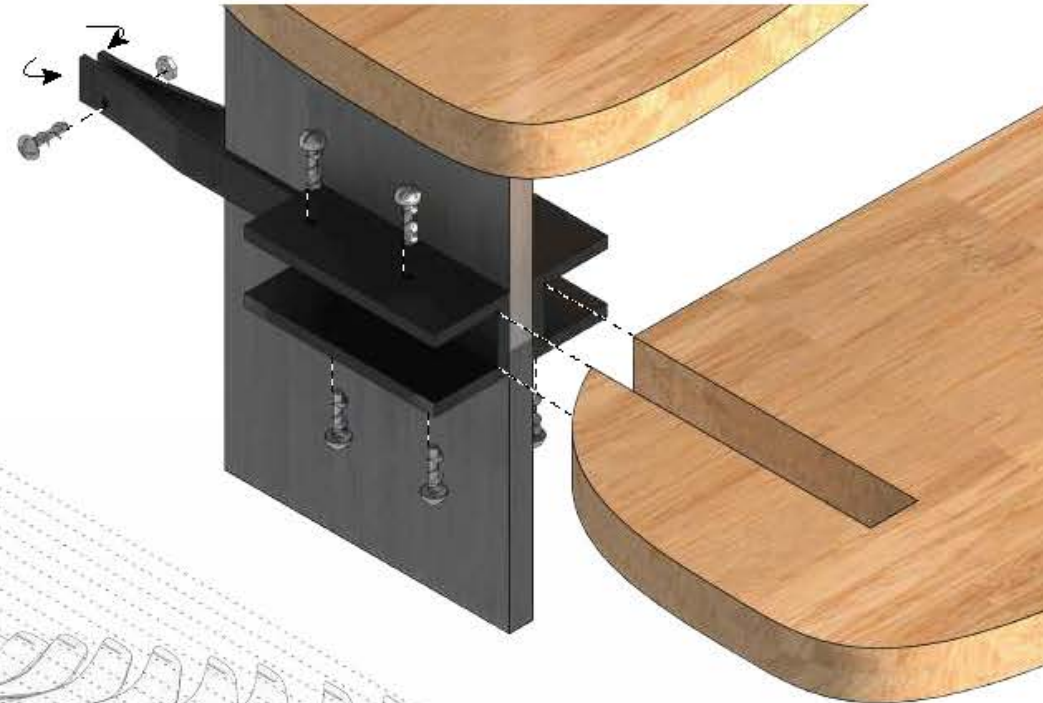
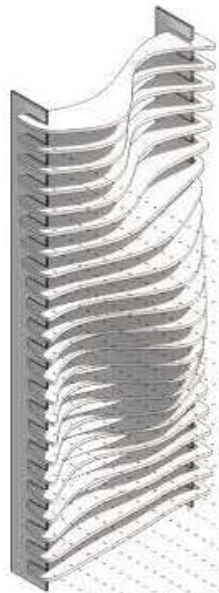
6 - screw



7 - fasten



metal joint assembly



lattice assembly

